Chapter 2

Overview of Wafer Contamination and Defectivity

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Chapter Outline

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2.1 WAFER CONTAMINATION

The criterion for a substance to be considered a contaminant is that it causes uncontrolled variations in the electrical performance of the device or in the device fabrication process. Contamination may originate from the wafer itself, from the cleanroom, from process tools, or from process chemicals and water. It may be observed visually or may only be detected with sophisticated analytical equipment during the inspection process or at the final device test.

According to the above definition, some (but not all) contaminants and defects have an impact on device performance. If the contamination causes a device to function improperly under predefined conditions, the contamination is said to have induced an electrical failure or defect. It is important to note that defects may also have other causes than contamination, such as poorly tuned process conditions or unexpected process variations. In this chapter, only defects that are related to contamination and defects that are unwillingly introduced by cleaning will be considered.

2.1.1 Classification of Contamination and Defects

It is possible to distinguish between groups of contaminants and defects that have similar behavior. Such a classification helps in organizing contamination control, but it should be noted that several classification schemes exist and every scheme has its limitations. This chapter will follow the conventional classification of contamination, which is largely based on detection methods. Also discussed in this chapter are several alternative classifications of contaminants.

2.1.1.1 Classification According to Detection Method

Many analytical tools are available to detect the presence of contamination on a wafer, and normally a contaminant can be detected with more than one type of analysis. For a review, refer to Chapters 11−13. The following types of contaminants are distinguished:

Metallic contamination: contamination consisting of (or containing) metallic species. Typically, for in-line monitoring no distinction is made in the speciation in which the contamination is present (i.e., an oxide, a silicide, a pure metal) even though this may influence the effect on devices. Common methods for detecting metallic contamination are total reflection X-ray fluorescence, atomic absorption spectrometry, and inductively coupled plasma−mass spectrometry (MS).

Particulate contamination: contamination that is usually observed by detection of light that is scattered off the defect on a surface. For this reason, a more correct name is "light-point defects." Thus, there is a need to distinguish between substrate defects that scatter light, such as scratches or crystal originated particles and actual material deposits. In this chapter, the term "particles" refers to the latter type of contamination.

Organic contamination: contamination containing carbon, plus the bonding structure associated with the *C*. One method for detecting surface contamination of this type is thermal desorption−MS, and other methods include X-ray photoelectron spectroscopy and Auger electron spectroscopy.

Surface defectivity: defectivity associated with the roughness of Si or film surface or the line edge roughness. Typically, optical methods or atomic force microscopy and associated techniques are used to detect these types of defects.

Atmospheric molecular contamination (AMC) and moisture: contamination that has molecular dimensions and therefore cannot be removed with normal high-efficiency particle air (HEPA) or ultra-low penetration air (ULPA) filters. AMC is usually monitored with ion mass spectrometry and capillary electrophoresis.

Some contaminants may fall into more than one category. For example, particles may consist of either organic or metallic material, or contain both, and thus arbitrarily be classified as particulate, metallic, or organic contamination. Fig. 2.1-1 shows the effects of metallic compound particles with a density of 5 g/cm^3 and a molecular mass of 100 amu (atomic mass unit) of the average metal concentration in atoms/cm^2 on a silicon wafer. As can be seen, a high concentration of metals on the wafer surface can result either from a large number of small particles or from a small number of large particles. As a result, to minimize particle-derived metal contamination on wafers, both the size and number of contaminant particles must be controlled.

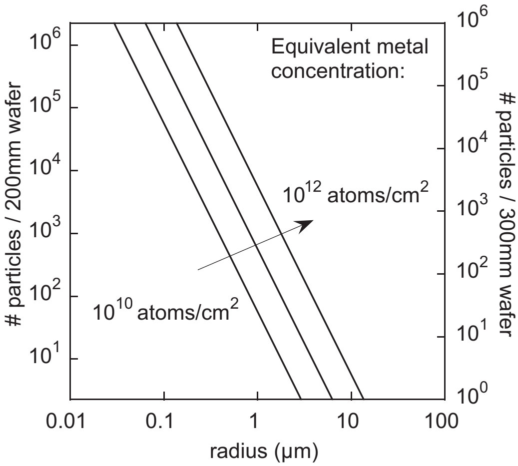

FIGURE 2.1-1 The effects of particle count and size on the equivalent metal concentration on a Si wafer surface. *Used with permission from authors.*

2.1.1.2 Classification According to Material

Technological progress has resulted in a diversification of materials that are used in the fabrication of silicon devices. Initially, the materials were limited to Si and the dopants necessary to control its semiconducting properties, SiO_2-based insulation layers, Si_3N_4 barriers, and sacrificial layers. For interconnect metallization, Al or Al/Cu alloys were used. Metal silicides were introduced to lower contact resistance. Later, TiN barriers and W plugs were used followed by Cu metallization and low-k (low dielectric constant) dielectrics, metallic oxides, and metal gate electrodes. While the use of these materials improves device performance, their uncontrolled presence in sensitive device regions can have a negative impact. Therefore, rather than being based on the detection method, a classification system may be based on the contaminant composition. In general, the contamination may be similar or dissimilar to the substrate material. This has important consequences for the cleaning method that is to be used to remove the contamination.

If the contamination consists of a different material than the substrate then it can, in principle, be removed by a material-selective cleaning method. Such methods may involve selective dissolution of the contaminant or selective dissolution of the substrate surrounding the contaminant. This "chemical cleaning" relies on the chemical properties of the substrate, the contaminant, and the cleaning agent. In practice, the material selectivity is finite, and side effects can occur that may in some instances be harmful. If the side effects become unacceptable, another cleaning agent must be found, or the chemical cleaning process must be enhanced or replaced by a physical cleaning force. As the wafer surface of the device becomes more complex, for example, when the surface is patterned and multiple materials are exposed or when gate-level features are exposed, the range of acceptable chemical cleaning options becomes more limited.

If the contamination consists of the same material as the substrate, the use of chemical cleaning agents is restricted, as it will not be possible to dissolve contaminants without dissolving substrate material. Physical cleaning methods, which are cleaning methods that employ physical forces for removal of contamination, become more important in such cases.

Purely mechanical cleaning methods that rely on the transfer of momentum from a cleaning medium such as an aqueous solution, a brush, or impinging particles of frozen gas, are not selective. The force they exert on the contaminant is also exerted on the wafer surface. In some cases, this can lead to damage of sensitive device structures.

Many current cleaning methods rely on a combination of chemical and physical methods. An example of such a combined method is undercut cleaning, in which the wafer surface surrounding an adherent contaminant particle is etched. As the particle-to-wafer contact area is reduced, repulsive electrostatic interactions between the particle and wafer become more important, until finally a condition is reached at which the electrostatic repulsion is strong enough to repel the particle from the surface [1–3]. Chapter 3 discusses this method in more detail.

2.1.1.3 Classification According to Force of Adhesion

The force of interaction between contamination and a substrate determines if, and how strongly, the contamination will adhere. The forces that play a role in this interaction can be used in a classification scheme. The interaction forces may be physical in nature, such as hydrophobic, van der Waals, and electrostatic forces. The interaction forces also may be chemical in nature, such as chemical bonding of contaminants on the substrate, or forces arising from the degree of hydration. Among these possibilities, chemical bonding interactions are the strongest. Contamination that is bound to a wafer through chemical bonds generally cannot be removed using physical methods unless the surface layer of the wafer is removed. Contamination that is bound to the wafer surface through hydrophobic, van der Waals, or electrostatic effects generally can be removed by both physical and chemical methods. Chapter 3 discusses the physics of adhesion.

2.1.1.4 Classification According to Size

Contamination on wafer surfaces may have characteristic dimensions that span several orders of magnitude with respect to the size of the particle or concentration of the contaminant. Molecular contamination involves both atoms and molecules. These contaminants may be physically adsorbed onto the wafer surface or may be chemically bound. For example, organic molecules from the cleanroom atmosphere may adsorb onto the wafer surface, while thin layers of oxidized Cu may result from the etching and post-etch cleaning of vias. Particulate contamination may have lengths or diameters ranging from nm to μm. Particles typically are not chemically bonded to wafer surfaces, but rather are held to the surface by a combination of hydrophobic, van der Waals, and electrostatic forces. Process-derived debris on the wafer surface, such as residues from etching or resist stripping processes, may have characteristic sizes ranging from nm to μm, as well. Most of the material debris is not chemically bonded either to the wafer or to itself, although the components of the debris in direct contact with the wafer surface may undergo bonding interactions. The ease of removal is based on a particle's size; the smaller the particle, the harder it is to remove. Size relationship to yield is discussed in Section 2.1.2.

2.1.1.5 Classification According to Yield and Reliability

Yield is defined as the fraction of functional devices at the end of the fabrication process. The defects causing device failure at this stage are often referred to as "killer defects." However, devices can also fail during their operational life due to "latent defects." Reliability is the ability of a device to perform its intended function over a certain period of time. Yield and reliability are often assumed to be correlated, the latent defect density being a small fraction of the killer defect density [4,5].

As in any manufacturing business the profitability is directly related to the cost of processing and to the number of devices that can be produced. At the same time, the performance of the chips is linked to their selling price. Failing devices can often be traced to uncontrolled or unwanted fluctuations in the fabrication process [6,7]. Contamination can be considered one such uncontrolled fluctuation. Cleaning of contamination from wafer surfaces, detection of contamination on wafer surfaces or in process systems, and determination of the cause of contamination and the method of transport to the wafer surface are therefore crucial to the optimization and stabilization of device fabrication.

Not every contaminant causes a defect that influences device performance [8]. This depends strongly on the contaminant properties such as its composition and location within the device, the design parameters such as the feature linewidths and pattern density in the vicinity of the contaminant, and the process and operating parameters such as the thermal budget following the introduction of the contaminant. Even if the optimum device performance is affected, the device may still meet its performance specifications [9–11]. The determination of "critical contamination levels" is therefore not straightforward.

2.1.2 Yield Models, Reliability, and Relationship to Defectivity

Models are used within the IC manufacturing community to relate defectivity to the device yield. Yield is defined as the fraction of devices that are functional at the end of the fabrication process. As stated in Section 2.1.1.5, not all particles and other contaminants lead to a decrease in yield and not all decreases in yield can be attributed to particles and contaminants. The defects causing device failure are often referred to as killer defects; these defects are detected during the final test process for the device. Latent defects refer to defects that cause the devices to fail during their operational lifetime, after the device has been tested, assembled, and packaged for consumer use. Finally, "reliability" is the ability of a device to perform its intended function over a certain period of time.

Since latent defect density and killer defect density are related it follows that their spatial distribution should also be correlated. That is, if killer defects are clustered, then latent defects are also expected to be clustered [12]. During device operation, the latent defects may grow to the critical defect size by Arrhenius growth, with the growth rate depending on factors, such as temperature or voltage [13−16]. The usefulness of such a defect growth model is limited to specific situations, because the reported activation energies show a significant spread. Moreover, the dominant failure mechanism under typical test conditions (high temperature, high voltage, and high moisture) may be different from the failure mechanism under operating conditions. Apart from the time dependence, reliability defects can be treated in a similar manner as killer defects.

Fundamental to most yield and reliability modeling is the assumption that the defects are small compared to the device area. Large defects such as scratches are typically not considered, possibly because they are relatively rare and usually not random.

2.1.2.1 Statistical Yield Models

Poisson distribution: This model defines yield as shown in Eq. (2.1-1):

$$\text{yield} = e^{-AD_0} \tag{2.1-1}$$

where A is the area of the device and D_0 is the defect density. The model assumes that defects occur randomly and independently from each other on the wafer surface. In practice this is not always true and some type of defect clustering occurs. If defects are clustered, then multiple defects, each of which is sufficient to produce chip failure, may be located on the same device. That is, for a given number of defects the actual yield value is normally higher than what is predicted by the Poisson model. To account for defect clustering, a weighting function $f(D)$ can be used:

$$\text{yield} = \int_0^{\infty} e^{-AD_0} f(D)\mathrm{d}D \tag{2.1-2}$$

For a discussion on different forms of $f(D)$ refer to Refs. [11,17,18].

Negative binomial yield model: An alternative to the Poisson model is the negative binomial yield model, as shown in Eq. (2.1-3):

$$\text{yield} = \left(1 + \frac{AD_0}{a}\right)^{-a} \tag{2.1-3}$$

The cluster factor a describes the clustering behavior of the defects; the probability of a defect occurring at a site (any specific location within the device) increases with the number of defects already present at that site. When $\alpha \rightarrow \infty$, the Poisson model is obtained. When $\alpha < \infty$ and for the same average defect density, the model gives higher yield values than the Poisson model. Often $\alpha = 2$ is chosen [11,18,19].

Weibull distribution: The Weibull distribution is widely used to describe the lifetime distributions of systems that fail due to the "weakest link." The cumulative probability function $F(t)$ is expressed as:

$$F(t) = 1 - \exp\left(-\frac{t-\gamma}{\eta}\right)^{\beta} \tag{2.1-4}$$

where γ is a location parameter, η is a scale parameter, β is a shape parameter, and t is the time. Depending on the value of β, a variety of behaviors can be described:

- $\beta < 1$: the failure rate decreases with time;
- $\beta = 1$: the failure rate is constant with time and the distribution is equal to the exponential distribution;
- $\beta > 1$: the failure rate increases with time.

In the absence of defects, the intrinsic device reliability dominates, $z(t)$, and the Weibull plot

$$z(t) = \ln[1 - \ln F(t)] \tag{2.1-5}$$

is a function of $\ln(t)$ and shows a straight line with slope β_i. If defects are present, both intrinsic and extrinsic device failures occur, and the Weibull distribution shows two straight lines for intrinsic and extrinsic failures with slopes β_i and β_e, respectively.

Using Eq. (2.1-1), and considering that

$$\text{yield} = 1 - F \tag{2.1-6}$$

it is shown that

$$\ln[-\ln(1 - F)] - \ln A = \ln D \tag{2.1-7}$$

where A is the device area and D is the defect density. This means that the Weibull distribution scales with device area and reliability predictions for small devices can be made on the basis of reliability measurements on large devices.

2.1.2.2 Defect Size Distribution

It is generally assumed that defects larger than a certain critical size cause device failures. Usually, this critical defect size is calculated as a fraction of the minimum feature size of the device [19,20]. As the minimum feature size shrinks, the critical defect size decreases as well. Unfortunately, small defects are more frequently observed than large defects, which means that the defect density D_0 increases as feature sizes are scaled down. Devices with small features are thus more vulnerable than devices with large features simply because there are more critically sized defects. It should be noted that in this context the defect size is defined as an "effective" electrical defect size [21]. This does not necessarily correspond to the physical particle size, since defects are not always particles.

To describe the defect size dependence, the defect density $D(d)$ is defined as a function of defect size d. Often a power law is used of the form

$$D(d) = D_0\left(\frac{d_0}{d}\right)^p \tag{2.1-8}$$

where D_0 is the defect density at a defect size $d > d_0$. This function breaks down near $d = 0$ and an alternative distribution function must be found. However, little is known about the impact and the distribution of such small defects, and the power law continues to be used down to the critical defect size [11]. If d_0 is chosen equal to the critical defect size, then D_0 represents the total density of critical defects. For the power parameter p, typically a value of about 2 is used [20,22,23], which corresponds to the aerosol size distribution in air. However, values as large as $p = 4$ have also been used [24], and it should be noted that the defect size distribution on the wafer is not necessarily the same as the aerosol size distribution in air [25].

Shrinking feature sizes by a factor B allows a reduction of the chip area by B^2. This has two consequences. First, the probability of failure is lower than if the chip area were constant. This partly compensates for the yield loss caused by the increase in defect density [24]. For example, assuming that $p = 2$ and that the critical defect size is proportional to the minimum feature size, the yield will remain constant. Second, the number of devices per wafer is increased. Even if the yield decreases with the implementation of smaller features, the total number of functioning devices per wafer may still be a greater number than without the shrink, with larger features [26].

2.1.2.3 Effective Area

Some areas of a device are more sensitive to defects than others. For this reason, an effective chip area $A_{eff}(d)$ can be defined that represents the area of the chip that is sensitive to defects of size d. For defects smaller than the critical defect size A_{eff} is assumed to be zero. For larger defects A_{eff} increases to a maximum that equals the device area. For devices containing many parallel lines an analytical expression for $A_{\text{eff}}(d)$ can be used [27]. Such equations are especially useful for chips containing regular arrays of features, such as memory devices. Other device types, such as microprocessors, exhibit a larger variation in feature size, and calculation of the effective area is more complicated. However, the larger features are relatively insensitive to defects. For more accurate estimates for A_{eff}, line lengths and device layout must be taken into account [28,29]. Using the effective area $A_{\text{eff}}(d)$ and the defect density $D(d)$ in the yield models described above, the defect density equation is

$$AD_0 = \int_0^{\infty} A_{\text{eff}}(d) \frac{\partial D(d)}{\partial d} \mathrm{d}d \tag{2.1-9}$$

where A is now an average effective value [24].

2.1.3 Mechanisms of Contamination

The process of wafer contamination may be described in terms of the mass and momentum transport processes that control the contaminant motion. According to the principle of mass conservation, the rate of change of the concentration of contaminant i in solution C_i is described by the gradient $\vec{\nabla}_i$ of the contaminant flux $\vec{J}_i$ and the net rate of generation/consumption S_i:

$$\frac{dC_i}{dt} + \vec{\nabla} \cdot \vec{J}_i = S_i \tag{2.1-10}$$

The contaminant flux $\vec{J}_i$ in any direction results from diffusion, convection, and the effects of an electric field on a charged species and may be described based on the Nernst–Planck equation:

$$\vec{J}_i = C_i \vec{u} - D_i \left[\vec{\nabla} C_i + z_i C_i \left(\frac{F}{RT} \right) \vec{\nabla} \phi \right] \tag{2.1-11}$$

where D_i is the diffusion coefficient of i in solution, z_i is the valence of species i, F is Faraday's constant, R is the ideal gas constant, ϕ is the local electrostatic potential, and $\vec{u}$ is the velocity vector of the solution [30]. When the flux to a wafer surface is desired, the vector components leading to transport to the wafer surface must be considered. The first term on the right-hand side of Eq. (2.1-11) describes the contaminant transport due to the convective flow of the solution. The second term describes the diffusion of i due to spatial variations in the concentration of i in solution, and the third term describes the motion of the contaminant in the electric field. At the interface with a substrate, at $x = x_0$, deposition and removal of contamination is in balance with the net flux:

$$J_i(x_0, t) = k_+ C_i(x_0, t) - k_- \sigma_{s,i}(t) \tag{2.1-12}$$

where k_+ and k_- are the characteristic rates for deposition and removal, and $\sigma_{s,i}$ is the concentration of contaminant i on the wafer surface. Note that in

Eq. (2.1-12), C_i has units of mole i per unit volume of solution, $\sigma_{s,i}$ has units of moles i per unit area of wafer surface, k_+ has units of length per time, and k_- has units of 1/time. Depending on the values for k_+, k_-, and the net flux J_i, the deposition/removal process can be either reaction rate limited or transport limited.

In the case of Eqs. (2.1-10)–(2.1-12), the term "reaction" refers to the partitioning of i onto the wafer from solution, whether by physical adsorption or by chemical reaction, and to the partitioning of i off the wafer surface into solution. If the contamination process is reaction rate limited, then the rate of transport of contamination to the wafer surface is fast compared to the rate of the partitioning of contamination onto the wafer. In such cases, the interactions between the contamination and wafer are of importance, including either chemical bonding interactions or chemical adsorption interactions. In certain cases, equilibrium between deposition and removal may be established, such that there is no net flux of contamination to the wafer surface, and the level of contamination on the wafer no longer changes with time.

If the contamination process is transport limited, then the rate of partitioning of contamination onto the wafer is fast compared to the rate at which the contamination is transported to the wafer surface. In such cases, the rate of contamination is dictated by the physical properties of the system, such as D_i, U_i, $\overrightarrow{u}$, and C_i. In the special case where diffusion is the main transport mechanism, the diffusion length L_{diff} describes the distance that contaminants can travel through the medium in time Δt:

$$L_{\text{diff},i} = \sqrt{2D_i\Delta t} \tag{2.1-13}$$

For a given change in $\sigma_{s,i}$, Eqs. (2.1-10)–(2.1-12) indicate that there will be a corresponding change in C_i. The ratio of $\Delta\sigma_{s,i}$ to ΔC_i defines a hypothetical equivalent layer thickness based on the contamination deposited on or removed from the surface:

$$L_{\text{ex},i} = -\frac{\Delta\sigma_{s,i}}{\Delta C_i} \tag{2.1-14}$$

Eqs. (2.1-13) and (2.1-14) are especially useful for describing contamination mechanisms. If $L_{\text{ex},i} < L_{\text{diff},i}$, then the exchange is reaction rate (deposition) limited. If $L_{\text{ex},i} \approx L_{\text{diff},i}$, then the exchange is diffusion limited. If $L_{\text{ex},i} > L_{\text{diff},i}$, then other transport mechanisms, such as convection, control the contamination process.

2.1.3.1 Contamination Transport Through Air

Cleanrooms are designed to ensure a continuous airflow that removes airborne contamination from sensitive areas such as the environment around the production equipment. The airflow in a cleanroom is often referred to as laminar, but it is actually turbulent. A more precise description is unidirectional. In the cleanroom it is important to avoid vortices or dead zones where the air is not refreshed (in corners, under tables, etc.) and where contaminants can accumulate. Perforated floor tiles, variable airflow velocities, and curtains can help to optimize the airflow. Placement of equipment can also have an impact, for example due to heat generation [31]. For this reason, modeling is an important aspect of cleanroom design.

2.1.3.2 Contamination Transport Through Liquids

Wafers come into contact with liquids during chemical mechanical polishing (CMP), and during cleaning, drying, and etching steps. In a static immersion tank $\overrightarrow{u} = 0$ (Eq. 2.1-11). The transport of contaminants occurs by diffusion and is driven by gradients in the contaminant concentration and gradients in the electrostatic potential. Over time, contamination will be distributed across the tank to yield a uniform contamination level. Thus, a wafer that contaminates the solution may eventually result in cross-contamination of other wafers [32].

If a laminar flow is present within the tank ($\overrightarrow{u} > 0$), contaminants are removed more efficiently, as the entire bath volume is smoothly swept by the fluid flow. If turbulence is present, the nonuniform velocity field may result in the creation of "dead spots" within the bath, which are not routinely drained by the flow and which may become reservoirs for contaminants. Due to the viscosity of water, a boundary layer exists near the wafer surface. In this region, the fluid velocity parallel to the wafer surface is assumed to vary from zero at the wafer surface to the bulk fluid velocity at the outer edge of the boundary layer. Using fluid dynamics, the thickness of the boundary layer δ is given by

$$\delta \sim 5\sqrt{\frac{\nu \cdot x}{u_0}} \tag{2.1-15}$$

with ν the kinematic viscosity, u_0 the liquid velocity away from the wafer, and x the position on the wafer, measured downstream from the edge. The boundary layer can be estimated to be several mm under typical processing conditions [33]. Within this layer, up to ~ 100 µm from the surface, the fluid velocity normal to the wafer surface can be assumed to be zero, and the contaminant transport is diffusion-limited [34]. As with static tanks, this fluid velocity can support both removal and redeposition of contamination [35]. The boundary layer thickness can be reduced, and hence the effect of convective contaminant removal enhanced, by the introduction of megasonic energy [36], resulting in more efficient removal of contamination. Wet cleaning methods that reduce the boundary layer are discussed in Chapter 4.

Other mechanisms for contaminant transport exist, depending on the specific situation. For example, during CMP contaminant transport to the wafer surface results from the slurry convection, contaminant diffusion, and electrophoresis, in addition to the applied force of the polishing pad against the wafer.

2.1.3.3 Basic Aspects of Wafer Surfaces

During front-end-of-line processing, typically two types of surfaces exist on a Si wafer: H-terminated Si and oxidized Si. The Pourbaix diagram of the Si H_2O system (Fig. 2.1-2) shows that in aqueous solutions Si forms stable, passivating oxide films, over the entire pH range. Immersion of Si in oxidizing solutions accelerates this process. The oxide thickness can vary between 0.6 and 1 nm depending on the solution. Because the oxidized surfaces become hydroxylated during exposure to air or water, they generally have a low contact angle. They are also easily contaminated with metallic impurities. Only at very high pH, >10, does the SiO_2 dissolve by forming soluble silicates.

Hydrogenated Si surfaces, as shown in Fig. 2.1-3, can be obtained by exposure to nonoxidizing environments, such as hydrogen fluoride (HF), caustic solutions with low oxidation potential (e.g., Si polishing), or H_2 ambient during high-temperature annealing. An advantage of H-passivated surfaces is that low nonnoble metallic contamination levels are easily obtained. However, noble metallic contamination is facilitated through charge transfer reactions at the interface, and due to their hydrophobic nature these surfaces are very sensitive to particle contamination. In addition, it is difficult to prevent the formation of water droplets and watermarks during drying of mixed hydrophobic/hydrophilic surfaces. Hydrogenated surfaces are unstable with respect to oxidation. The oxidation by O_2 is typically catalyzed by H_2O.

Oxidized Si is usually hydroxylated and the surface is terminated by silanol groups (Si—OH). These polar groups will form H-bonds with neighboring H_2O

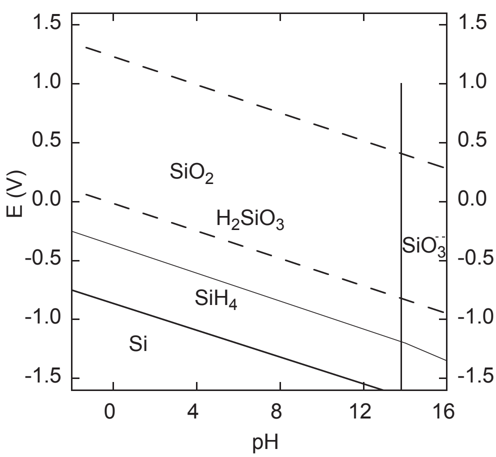

FIGURE 2.1-2 Potential—pH equilibrium diagram for the system Si—H_2O at 25°C and 1 bar, showing the regions of stability for Si and SiO_2 and the predominance areas of ions in aqueous solution. *Based on data in M. Pourbaix, Atlas of Electrochemical Equilibria in Aqueous Solutions, National Association of Corrosion Engineers, Houston, Texas, 1974.*

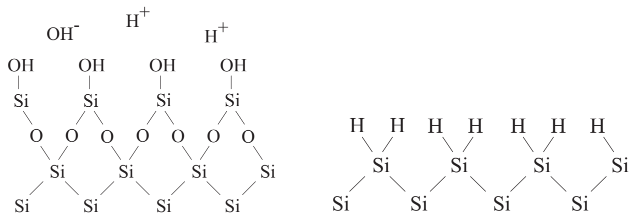

FIGURE 2.1-3 Schematic representation of a hydroxylated/hydrophilic surface (left) and a hydrogenated/hydrophobic surface (right). *Used with permission from the authors.*

molecules (hydration). In aqueous solutions, silanol groups can also act as a weak base or acid [37,38] according to

$$\begin{aligned} &\equiv \mathrm{SiOH} + \mathrm{H}^+ \leftrightarrow \equiv \mathrm{SiOH}_2^+ \quad \mathrm{p}K_+ = -1.9 \\ &\equiv \mathrm{SiOH} \leftrightarrow \equiv \mathrm{SiO}^- + \mathrm{H}^+ \quad \mathrm{p}K_- = -7 \end{aligned} \tag{2.1-16}$$

with K_+ and K_- the protonation and deprotonation constants. Due to this protonation/deprotonation behavior the silicon surface will bear a negative or positive surface charge σ_0 [39]:

$$\sigma_0 = qN_0 \frac{\left[\mathrm{SiOH}_2^+\right] - \left[\mathrm{SiO}^-\right]}{\left[\mathrm{SiOH}_2^+\right] + [\mathrm{SiOH}] + \left[\mathrm{SiO}^-\right]} \tag{2.1-17}$$

where q is the elementary charge and N_0 the surface density of adsorption sites. The point of zero charge (PZC) is defined as the pH of the solution where the surface bears no net charge, i.e., for Si $\left[\mathrm{SiOH}_2^+\right] = \left[\mathrm{SiO}^-\right]$:

$$p\mathrm{H}_{\mathrm{PZC}} = \frac{pK_+ - pK_-}{2} \tag{2.1-18}$$

for Si, $p\mathrm{H}_{\mathrm{PZC}} = 2.5$.

Ions in solution can dehydrate and specifically adsorb to charged surface sites, forming a sheet of charge, which is called the inner Helmholtz plane (IHP). Alternative names for this layer are Stern layer, Helmholtz layer, and compact layer. The charges are balanced by a charge redistribution on the silicon and in the liquid. In the liquid, this diffuse layer starts at the outer Helmholtz plane (OHP), which is typically one monolayer of water from the IHP, due to the ions being hydrated. Beyond the OHP, the concentration of ions varies exponentially with electrostatic potential φ until both reach their equilibrium in the bulk of the solution. For example, the concentration of H^+ ions can be written as

$$[H^+] = [H^+]_{\mathrm{bulk}} \exp\left[-\frac{q\varphi}{kT}\right] \tag{2.1-19}$$

Using the Poisson equation, and assuming that only monovalent ions are present, the integrated charge in the diffuse layer per unit area is

$$\sigma_d = \frac{2\varepsilon}{L_D}\frac{kT}{q}\sinh\left[\frac{q\varphi_d}{2kT}\right] \quad \text{with} \quad L_D = \sqrt{\frac{kT/q}{qC/\varepsilon}} \tag{2.1-20}$$

with ϕ_d the potential at the OHP and C the concentration of ions in solution, and T and k are the temperature and Boltzmann's constant, respectively. It should be noted that the dielectric constant ε close to the surface deviates from the dielectric constant of bulk H_2O due to the polarization of H_2O molecules. The constant L_D is the Debye−Huckel length, which is measured for the range over which electrostatic interactions occur. A very similar analysis can be done for the charge redistribution in the silicon [40], resulting in a Debye length:

$$L_D = \sqrt{\frac{2kT/q}{q(n+p)/\varepsilon}} \tag{2.1-21}$$

where n and p are the electron and hole densities, respectively. The expression for the charge σ_s in the silicon depends on its sign and magnitude, and will not be given here. However, it is clear that the interaction of the Si surface with contaminants depends not only on the pH of the solution and its ionic strength, but also on the substrate doping type and concentration. The number of adsorption sites for Si substrates is reported to vary in the range $10^{12}-10^{13}\ \mathrm{cm}^{-2}$, but this also depends strongly on preparation method [38,41,42]. The interactions described here are not restricted to Si, but also occur at other surfaces such as silica, Si_3N_4, and Al_2O_3.

2.2 BEHAVIOR AND IMPACT OF CONTAMINATION

2.2.1 Particle Contamination

2.2.1.1 Origins of Particle Contamination

Particles are always present in the atmosphere. Outside cleanrooms, particulate contamination may be visible as smog, fog, or dust, that have varying origins, such as emissions from human activities, desert sand, volcanic ash, and moisture. The particle size may vary over a wide range, from less than 10 nm to over 100 μm. Moreover, large particles may break up into smaller particles, while small particles may agglomerate to form larger ones. The following categories of particles are usually distinguished:

Nucleation mode particles with diameters less than 100 nm are mainly formed by condensation of hot gases from combustion sources (e.g., diesel motors), sulfate, ammonia, and volatile organic compounds. They can reach high concentrations, especially in urban areas.

Accumulation mode particles are approximately 0.1−1 μm in diameter. These particles represent the largest fraction of the total surface area in an aerosol. Nucleation mode particles and gaseous species can adsorb on this area, causing the particle to grow.

"*Coarse mode*" *particles* are larger than 1 μm in diameter, and originate from mechanical processes (wear), soil, and dust. Examples of sources of wear-generated particles in a cleanroom are ball bearings, wafers sliding in carriers, and wafer clamping mechanisms, among others [43]. While accumulation mode particles may grow to this size, this is not likely to occur in a cleanroom environment.

In aerosol science, experimental particle size distributions often closely follow a lognormal size distribution. However, it has also been observed that in many cases a power law with parameter a can approximate particle size distributions:

$$\mathrm{d}N = f(d)\mathrm{d}d \quad \text{with} \quad f(d) = \frac{C_0}{ad_0}\left(\frac{d_0}{d}\right)^a$$
$$\text{or} \qquad C = C_0\left(\frac{d_0}{d}\right)^{a-1} \tag{2.2-22}$$

where C is the concentration of particles with a diameter larger than d. It is observed that usually $\alpha \approx 3$. Assuming that the particles are spherical, this observation implies that the total particle volume, V, for each incremental change in diameter, d, remains constant:

$$\mathrm{d}V = \frac{1}{6}\pi d^3 \mathrm{d}N \tag{2.2-23}$$

Fig. 2.2-4 shows an application for monitoring air quality. The different categories of particles clearly can be discerned.

The power law size distribution is adopted by standards describing cleanroom class specifications. The current standard ISO 14644-1 (International Standards Organization) assumes the following particle size distribution function [22]:

$$C_m = 10^m\left(\frac{0.1}{d}\right)^{2.08} \tag{2.2-24}$$

where C_m is the number of particles per m^3 with diameter d in μm. The reference diameter is 0.1 μm. The factor m denotes the cleanroom classification number and is determined from Eq. (2.2-23). The ISO standard replaced the (US) Federal Standard 209 [23], the terminology of which is more widespread:

$$C_n = 35.31n\left(\frac{0.5}{d}\right)^{2.2} \tag{2.2-25}$$

where n is the allowed number of particles per ft^3 at 0.5 μm particle diameter. The factor 35.31 converts the US standard to SI units (particles per m^3).

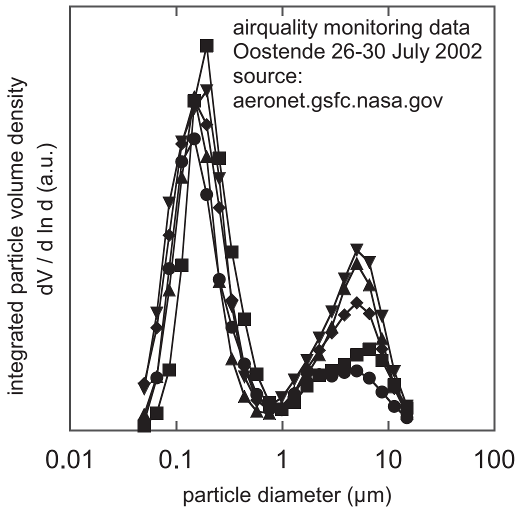

FIGURE 2.2-4 Example of particle volume density as a function of particle diameter [44]. *Used with permission from aeronet.gsfc.nasa.gov and MUMM/RBINS.*

A comparison of the ISO standard and the Federal Standard 209 is shown in Fig. 2.2-5, along with actual data obtained from a class 1000 cleanroom. For particles smaller than 300 nm, the particle count does not obey the power law, Eq. (2.2-23).

Typically, two types of filters are used in the cleanroom. HEPA filters are rated to 99.99% efficiency for particles 300 nm and larger in diameter and ULPA filters are rated to 99.999% efficiency for particles 120 nm in diameter.

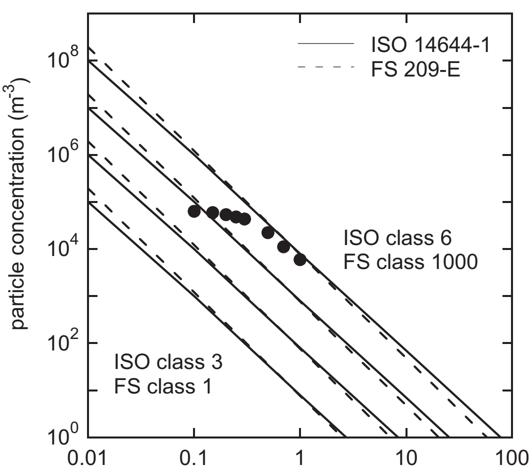

FIGURE 2.2-5 Comparison of different cleanroom classes. Also shown are actual monitoring data from a class 1000 cleanroom. *Used with permission from the authors.*

HEPA filters have minimum efficiency at 300–400 nm, smaller particles are more easily removed because they exhibit significant Brownian motion and are more easily captured by the filter.

2.2.1.2 Effects of Particulate Contamination

The detrimental impact of particles is one of the reasons microelectronic devices are fabricated in the controlled environment of a cleanroom. Particle count is one of the most important criteria for process control. At the same time, there is surprisingly little experimental evidence for the detrimental impact of particles. In the following sections, the effects of particles present on the front and back surfaces of wafers are discussed.

2.2.1.2.1 Front Surface Particles

The most prominent area of concern is related to lithographic patterning. The presence of front surface particles distorts the pattern on the wafer. The critical particle diameter is the diameter above which a particle becomes a "killer defect," defined as a defect that affects the functionality of the device. Usually the critical particle diameter is determined to be ½ the minimum feature size [19], but there is little experimental evidence to support this. Intuitively, for technology that is at its limits of the definition of small patterns, the ½ feature size criterion should be more than sufficient. However, theoretical treatments of 248-nm lithography indicate that a ¼ feature size criterion, depending on the location and the composition of the particles [45,46], may be more appropriate.

The possible detrimental effects of front surface particles are not restricted to distortion of the lithographic pattern. Local masking could occur during (anisotropic) etching or ion implantation. Many defect impact studies have focused on back-end-of-line processing [4,25,47], because of the relatively high pattern densities and because of the availability of simple test devices such as meanders and forks. As a result, opens and shorts are reported as the most dominant defects. The particle diameter is often comparable to, or much larger than, the layer thickness that is used for processing. For this reason it is anticipated that particles have an effect on the properties of device layers [48]. The impact may be a function of particle material and size, and layer thickness, composition, and deposition method.

2.2.1.2.2 Back Surface Particles

The impact of back surface particle contamination is difficult to determine when processing single-side polished wafers. The main reason is that the roughness of the wafer back surface makes optical detection difficult except for very large particles. Moreover, it is generally believed that small particles can hide in the surface roughness of the back surface and therefore have limited effect. The use of double-side polished wafers has significantly improved the detection limits for back surface particles and simultaneously has increased the concern. Additionally, if particles that originate on the substrate back surface are transferred to a wafer chuck or electrode in a processing tool, they may cause systematic yield loss for the many wafers processed after the contaminated wafer. However, systematic studies on the effects of back surface particle contamination are sparse [49].

Back surface particles are reported to be a source for front surface particle contamination by "fall-on" from the back surface of one wafer that is adjacent to the front surface of another wafer [50,51]. However, when comparing the gravitational force and the van der Waals adhesion force, as discussed in Chapter 3, it readily is seen that the "fall-on" mechanism is only likely for particles larger than ~100 μm. Another mechanism for yield loss could be a degraded contact from electrostatic chucks [50].

An often-quoted effect of back surface particles is the formation of "hot spots" during lithographic pattern definition [50,52,53]. The vacuum that holds the wafer on the chuck causes a local deformation of the chuck, particle, and wafer due to the presence of the particle. The local forces are so large that a plastic deformation of the particle, chuck, or substrate occurs [49]. With particles larger than 10 μm, the deformation can be large enough to cause focusing problems during lithographic exposure. Fig. 2.2-6 shows the deformation of particles. During CMP, similar effects may be anticipated. It should be noted that "hot-spots" can be several mm in diameter, and can therefore no longer be considered as point defects, because there is a finite probability that one defect can affect several devices.

The impact of back surface particles often is not determined by its morphology or size, but by its composition. The particles may contain metallic

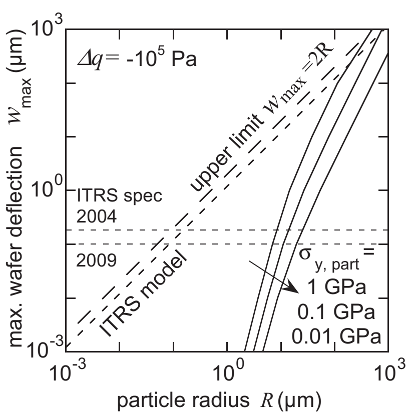

FIGURE 2.2-6 Maximum wafer deflection as a function of particle radius, assuming plastic particle deformation. Different values for the particle yield strength $\sigma_{y,part}$ are shown [49]. *Reprinted with permission from T. Bearda, P.W. Mertens, F. Holsteyns, P. De Bisschop, R. Compen, A. van Meer, M.M. Heyns, Jpn. J. Appl. Phy. 44 (2005) 7409.*

species that can diffuse into the wafer and affect the electrical performance of the devices. Tools employing back surface handling can acquire contaminants and pass them onto wafers, which in turn contaminate other tools and wafers. The feasibility of this mechanism is illustrated by common practice: cycling a number of dummy wafers through the handling system can reduce handling-induced contamination.

Back surface contamination occurs not only in the form of particles; both metallic and organic contaminants are also detrimental. Copper, in particular, must be removed from the wafer back surface and edge bevel to avoid the necessity of using dedicated tool sets.

2.2.1.2.3 Particles on Lithographic Media

Particulate contamination present on mask surfaces can cause defects to print onto the wafer patterned with the contaminated masks, especially if a mask without a pellicle is used. As the wavelength of light used to pattern devices shrinks, defects on the masks that are nm-scale will print defective devices, while adherent particle contaminants will act as defects [19].

2.2.2 Metallic Contamination

Wafers may become contaminated with metals as particles or in molecular or atomic states. Both types of contamination often occur in a liquid medium, such as a cleaning or etching mixture. As discussed in Chapter 9, the properties of the substrate, for example a Si surface with a native oxide, are important. Also important are the composition of the aqueous solution, the chemical processes occurring within the solution and on the substrate surface, and the composition of the metallic contamination in the aqueous solution.

2.2.2.1 Metallic Contamination From Aqueous Media

Pourbaix diagrams provide information about the stable speciation of metals in aqueous media. The diagrams are derived for specific ionic concentrations, pressures, and temperatures. They represent a thermodynamic equilibrium, but do not take into account kinetic behavior. For example, Fig. 2.2-7 shows the Pourbaix diagram of Ca in H_2O and shows that trace amounts of Ca can be expected to be present as soluble Ca^{2+}. In typical cases, metals are present as cations at low pH and as hydroxides at high pH.

The use of Pourbaix diagrams is subject to the condition that complexing species are not present and insoluble salts are not formed. Obviously this is not always true. Table 2.2-1 lists the solubilities of some sparingly soluble metal hydroxides and metal fluorides. These salts will precipitate if their concentration is high enough. This is especially true for the case of hydroxide salts at high pH [55,56], and also true for certain fluoride salts. Their solubilities, at 2 wt% (weight %) HF, seem quite high compared to typical levels in cleaning mixtures; however, the solubility may be, in some cases, an issue for

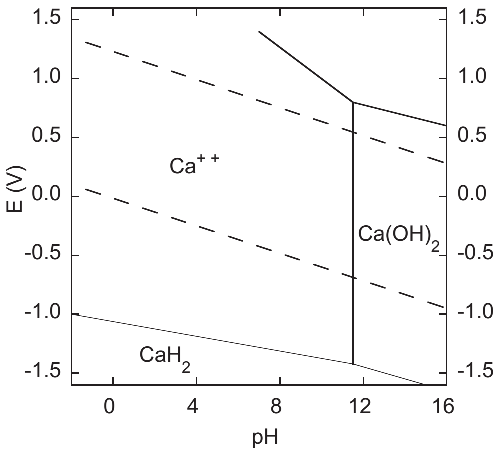

FIGURE 2.2-7 Pourbaix diagram of Ca in water at 25°C. The dotted lines enclose the region of stability for water. *Based on data in M. Pourbaix, Atlas of Electrochemical Equilibria in Aqueous Solutions, National Association of Corrosion Engineers, Houston, Texas, 1974.*

contamination analysis when the contaminant is collected using the vapor phase decomposition technique, as discussed in Chapter 13.

A metal complex is a metal ion surrounded by other molecules called ligands. Most metal ions in water actually exist as water complexes, such as $Mt(H_2O)_6^{2+}$ or $Mt(H_2O)_6^{3+}$, where Mt represents the metal ion. Depending on the chemistry, the H_2O molecules can be replaced by other ligands such as F^-, Cl^-, and NH_3^+. A list of complexes and their formation constants is shown in Table 2.2-2. Some practical implications of complex formation include lower Cu, Ni, and Zn contamination levels (compared to other contaminants) when a substrate is immersed in an aqueous solution containing ammonia or the formation of $CuCl_2^-$ in HF/HCl mixtures, which prevents electrochemical reduction of the Cu^{2+} at Si surfaces.

2.2.2.2 Metallic Ion Exchange Reactions

When SiO_x is exposed to an aqueous chemistry, an equilibrium is established between protonated and deprotonated silanol (SiOH) groups. The effect on surface charge has been discussed previously in Section 2.2.1.3. When metallic cations are present in the solution, they not only will be affected by the surface charge, but they also will participate in the surface reaction:

$$\equiv\mathrm{SiOH(s)} + \mathrm{Mt}_i^{n+}\mathrm{(aq)} \leftrightarrow \equiv\mathrm{SiOMt}_i^{(n-1)+}\mathrm{(s)} + \mathrm{H^+(aq)} \tag{2.2-26}$$

with an associated reaction rate constant K_{Mi}. In this case, the adsorption process is referred to as chemisorption, as the metal ions chemically react with the surface upon adsorption. It should be noted that the aqueous H^+ species would typically exist as H_3O^+ in solution. Metal ions are in competition with

TABLE 2.2-1 Solubility Products of Sparingly Soluble Metal Hydroxides and Fluorides [64]

Metal	Hydroxides Formula	K_{sp}	Maximum Dissolved pH = 2 (ppm)	pH = 7	pH = 10	Fluorides Formula	K^{sp}	Maximum Dissolved 2% HF
Al	$Al(OH)_3$	3.00×10^{-34}	>1000	<1 ppt	<1 ppt			
Ca	$Ca(OH)_2$	5.02×10^{-06}	>1000	>1000 ppm	2 ppb	CaF_2	3.45E-11	2 ppb
Co	$Co(OH)_2$	5.92×10^{-15}	>1000	>1000 ppm	35 ppb			
Cr	$Cr(OH)_3$	6.30×10^{-31}	>1000	<1 ppt	<1 ppt	CrF_3	6.60E-11	200 ppb
Cu	$Cu(OH)_2$	4.80×10^{-20}	>1000	300 ppb	<1 ppt			
Fe	$Fe(OH)_2$	4.87×10^{-17}	>1000	300 ppm	300 ppt	FeF_2	2.36E-6	200 ppm
Fe	$Fe(OH)_3$	2.79×10^{-39}	160	<1 ppt	<1 ppt			
Li						LiF	1.84E-3	500 ppm
Mg	$Mg(OH)_2$	5.61×10^{-12}	>1000	>1000 ppm	15 ppm	MgF_2	5.16E-11	2 ppb
Mn	$Mn(OH)_2$	2.00×10^{-13}	>1000	>1000 ppm	1 ppm			
Ni	$Ni(OH)_2$	5.48×10^{-16}	>1000	>1000 ppm	3 ppb			
Zn	$Zn(OH)_2$	3.00×10^{-17}	>1000	200 ppm	<1 ppt	ZnF_2	3.04E-2	>1000 ppm

Included are calculated maximum levels for some typical conditions (compiled by authors). For HF, a dissociation constant of 7.2×10^{-4} was used.
Used with permission from CRC Press Inc.

TABLE 2.2-2 Overview of Metal Complexes and Their Formation Constants [54]

Metal	Hydroxide	Log *K*	Ammonia, Fluoride, and Chloride	Log *K*
Al	$[Al(OH)_2]^-$	18.6	$[AlF_6]^{3-}$	19.8
Co	$[Co(OH)]^{2+}$	5.2	$[Co(NH_3)_6]^{3+}$	4.2
Co	$[Co(OH)_4]^{2-}$	5.4	$[Co(NH_3)_6]^{2+}$	4.4
Cr	$[Cr(OH)_2]^+$	5.4	$[CrF_3]$	10.2
Cu	$[Cu(OH)_4]^{2-}$	5.6	$[CuCl_2]^-$	5.5
Cu			$[Cu(NH_3)_2]^+$	10.6
Cu			$[Cu(NH_3)_4]^{2+}$	11.8
Fe	$[Fe(OH)_4]^-$	7.7	$[FeF_3]$	11.9
Fe	$[Fe(OH)_4]^{2-}$	9.6		
Mn	$[Mn(OH)]^{2+}$	11.7	$[MnF]^{2+}$	5.7
Mn	$[Mn(OH)_4]^{2-}$	17.3		
Ni	$[Ni(OH)_3]^-$	14.4	$[Ni(NH_3)_6]^{2+}$	8.3
Ti	$[Ti(OH)]^{2+}$	52.8		
Ti	$[Ti(OH)_4]^{2-}$	18.3		
V	$[V(OH)]^{2+}$	25.8		
V	$[VO(OH)_2]$	18.0	$[VO_2F_4]^{3-}$	7.0
Zn	$[Zn(OH)_4]^{2-}$	33.9	$[Zn(NH_3)_4]^{2+}$	8.9

The selection was made based on the criterion that more than 0.01 wt% of the metal ions would be complexed with a ligand concentration of 1.0 M and a total metal concentration of 1.0×10^{-8} M.
R.M. Smith, A.E. Martell, Critical Stability Constants, in: Inorganic Complexes, vol. 4, Plenum Press, New York, London, 1976. Reprinted with kind permission of Springer and Business Media.

H^+ for adsorption on the surface and will in turn influence the surface charge. The adsorption is in balance with the energy required to disrupt cation and surface hydration, and with the electrostatic interaction between the cation and the surface.

In a simplified case, in which all metallic cations are monovalent, the surface concentration N_{SiOMt_i} of a metal Mt_i can be calculated based on Eq. (2.2-26) [42]:

$$\frac{N_{SiOMt_i}}{N_0} = \frac{K_{M_i}\left[Mt_i^+\right]e^{-q\varphi_d/kT}}{1 + K_+[H^+]e^{-q\varphi_d/kT} + K_-[H^+]^{-1}e^{q\phi_d/kT} + \sum K_{Mt_i}\left[Mt_i^+\right]e^{-q\phi_d/kT}} \tag{2.2-27}$$

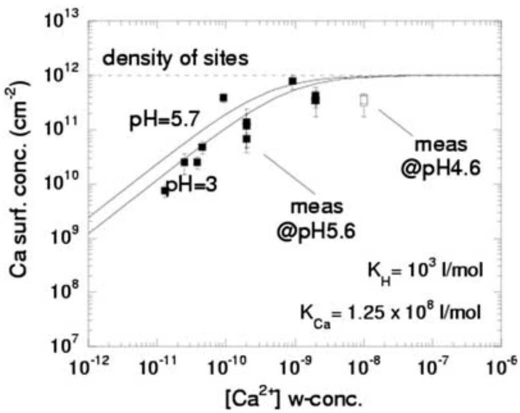

FIGURE 2.2-8 Experimental results [57] and the model of Eq. (2.2-26) for N_{Ca}/Nt as a function of Ca^{2+} weight–concentration. *Used with permission from Semiconductor FabTech.*

The parameter values can be obtained by fitting experimental data to Eq. (2.2-27). Fig. 2.2-8 shows the agreement between experimental data [57] and the model in a conventional deposition plot.

2.2.2.3 Metallic Redox Reactions

When the Si surface is H-terminated, metals that are more noble than Si, such as Au, Ag, Cu, Pt, Ir, Ru, and Pd, can participate in a redox reaction at the silicon surface, and deposit on the surface. The reduction of the noble metals can take place by either the release of conduction band electrons or the injection of valence band holes. The reaction rate is determined by the overlap between occupied electron energy states at the silicon surface and unoccupied electron energy states of the metal redox system. The reaction therefore strongly depends on substrate doping type and level.

Fig. 2.2-9 shows schematically two different metal redox systems M_1 and M_2 with their respective redox levels E^0_{M1} and E^0_{M2}. Fig. 2.2-10 shows the position of the conduction and valence band edge of the Si substrate in contact with the dilute HF solution and the redox potential of the different noble metal redox systems.

As an example, the reduction of Cu^{2+} ions in two consecutive charge transfer steps is shown:

$$\begin{aligned} Cu^{2+} + e^- &\rightarrow Cu^+, \quad E^0 = 0.153\ \mathrm{V} \\ Cu^+ + e^- &\rightarrow Cu, \quad\ \ E^0 = 0.521\ \mathrm{V} \end{aligned} \tag{2.2-28}$$

Because the redox potential $E^0_{Cu^{2+}/Cu^+}$ lies well above the valence band edge of Si, the reduction of Cu^{2+} will primarily proceed over the conduction band. The reduction of Cu^+ occurs by valence band hole injection. Holes are

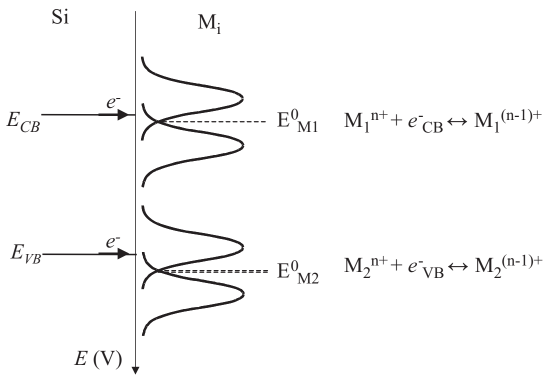

FIGURE 2.2-9 Comparison of the electrochemical electron energy levels in a semiconductor and the unoccupied and occupied electron energy states of two different redox systems $M_1^{n+}/M_1^{(n-1)+}$ and $M_2^{n+}/M_2^{(n-1)+}$ with E^0_{M1} and E^0_{M2}, their respective standard redox potential [58]. *Used with permission from Ivo Teerlinck and University of Ghent.*

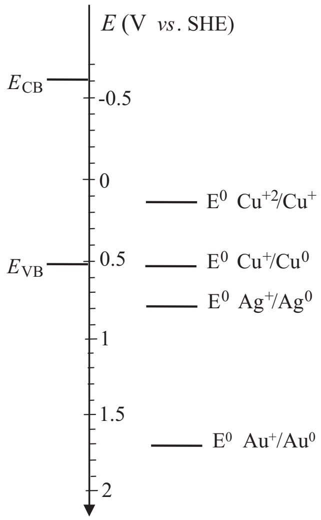

FIGURE 2.2-10 Energy level diagram showing the position of the redox Fermi levels associated with Cu, Ag, and Au relative to the band-edges of Si in 1 M HF solution [58]. *Used with permission from Ivo Teerlinck and University of Ghent.*

also required for the oxidation of silicon. Because both types of charge carriers take part in the reaction, the supply of minority carriers to the surface will be rate limiting. This explains the strong dependence on the illumination conditions: in the dark, the minority carrier concentrations are low resulting in little Cu outplating and Si etching. Under illumination, however, electron-hole pairs are generated. In p-type Si, photogenerated electrons are used for the reduction

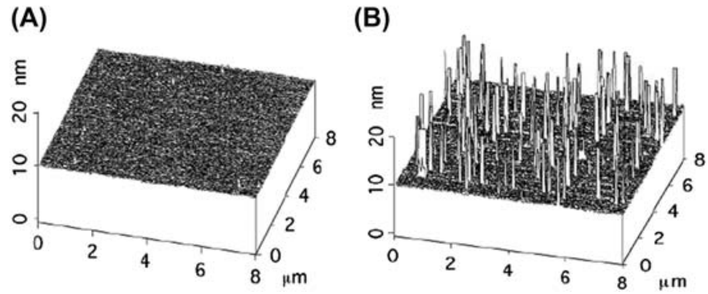

FIGURE 2.2-11 Atomic force microscopy (AFM) plots of a p-type Si wafer surface after 10 min immersion in a 0.5% HF solution containing 100 ppb Cu ions. The immersion was performed in (A) the dark or (B) under illumination [58]. *Used with permission from Ivo Teerlinck and University of Ghent.*

of the Cu^{2+} ions, while for n-type Si photogenerated holes are consumed by the Si oxidation. Thus for both p- and n-type Si wafers, the Cu outplating is enhanced under illumination, as shown in Fig. 2.2-11 [58]. Measures can be taken, such as adding complexing agents to the solution, to prevent outplating.

For redox systems with a more positive reduction potential, such as Ag^+/Ag or Au^+/Au, reduction by valence band hole injection becomes more likely. Since the deposition is not limited by the minority carrier concentration, the deposition kinetics become independent of the illumination conditions and metal deposition is expected in darkness, when compared to Cu system.

The electrochemical oxidation of Si in HF results in a rough surface [59–61]. The exact mechanism behind this is not completely understood and depends on many parameters such as redox potentials, Fermi levels, and charge carrier density in the silicon substrate. Some evidence exists that the oxidation of Si proceeds according to:

$$Si + 2HF + 2H^+ \leftrightarrow SiF_2 + 2H^+ \tag{2.2-29}$$

The unstable SiF_2 reacts further with HF and/or H_2O molecules to form SiF_6^{2-}.

2.2.2.4 Behavior of Metallic Contamination

Whereas particulate contamination is often considered "inert" during device processing, metallic contamination is highly reactive. Depending on the temperature and the ambient atmosphere, metal compounds such as metal oxides, silicates, and silicides may be formed [62]. Alternatively, metallic contaminants can dissolve into the Si substrate crystal structure, or can evaporate, or sublimate from the surface. An example is the behavior of Fe contamination during thermal oxidation [63]; if the temperature ramp-up is performed in an ambient atmosphere containing O_2, Fe is immobilized at the wafer surface. If, on the other hand, the ramp-up ambient atmosphere is pure N_2, Fe diffuses into the Si substrate. The behavior thus strongly depends on the contaminant, the substrate material, and the process used. The heat of formation of several metal oxides and silicides can be found in Ref. [64]. The following sections describe the behavior of metals in Si and dielectric materials.

2.2.2.4.1 Metals in Silicon

Metallic impurities may dissolve in the Si and diffuse to sensitive device areas [65,66]. Fig. 2.2-12 shows the diffusivities of numerous metals in Si. Among the metals with the highest diffusivities, the transition metals (Cu, Fe, Ni, Co, Cr, and Mn) are most relevant for processing of silicon devices. Other metals (Al, group II metals, Ti, and Sc) may be relevant, but have much lower diffusivities and therefore usually will not diffuse significantly into the silicon.

Many impurities do not dissolve as neutral species, but are present as charged ions. This means that the dopant type and concentration in the Si influences the behavior of the impurities, both directly by complex formation and indirectly by interaction with the electrostatic potential (Fermi level). Well documented is the pairing of Fe^{+} and Cu^{+} with negatively charged B atoms in p-type Si [67,68]; additionally, Mn—B and Co—B complexes also are known to form [66]. Such interactions reduce the effective diffusivities of the impurities and increase their solubilities.

Metals in Si have a finite solubility that decreases with decreasing temperature, as shown in Fig. 2.2-13. If the metal concentration exceeds the solubility, the dissolved metals will precipitate, usually by silicide formation. Energy is needed for precipitation due to the volume expansion; for example,

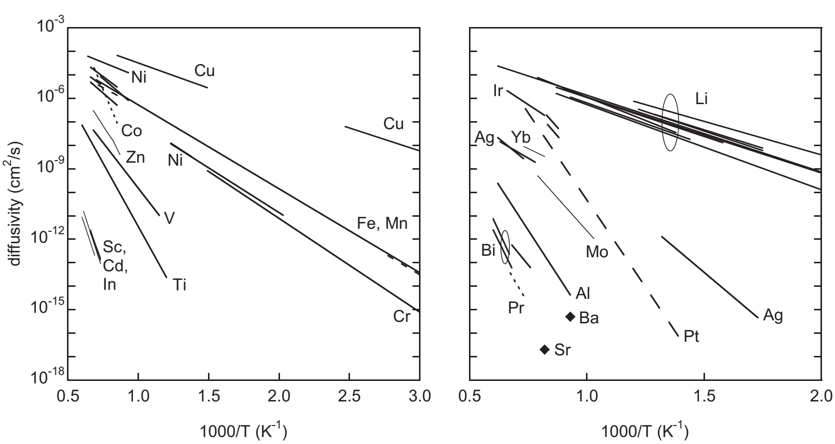

FIGURE 2.2-12 Diffusivities of metals in Si. *Compiled by the authors from data found in F.H. Wöhlbier, Diffusion and Defect Data, Solid State Data, vol. 47, Trans Tech Publications Inc., Aedermannsdorf, 1986; D.J. Fisher (Ed.), Diffusion in Silicon, 10 Years of Research, Trans Tech Publications Inc., 1998; W. Schröter, M. Seibt, in: R. Hull (Ed.), Solubility and Diffusion of Transition Metal Impurities in c-Si Properties of Silicon, INSPEC, London, 1999, p. 543.*

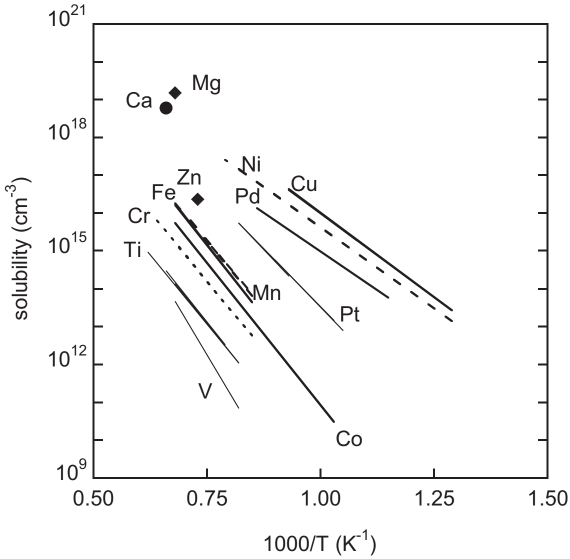

FIGURE 2.2-13 Solubilities of various metals in Si. *Compiled by the authors from data in W. Schröter, M. Seibt, in: R. Hull (Ed.), Solubility and Diffusion of Transition Metal Impurities in c-Si Properties of Silicon, INSPEC, London, 1999, p. 543.*

during the formation of Cu_3Si the volume increase is estimated to be a factor of 2–3 [69]. If many Si self-interstitial sites are present (Si atoms that are positioned at off-lattice sites in the crystal matrix), then precipitation is reduced [70]. In addition, Cu_3Si precipitates carry a charge that depends on the Fermi level of the substrate, resulting in electrostatic interaction with positively charged interstitial Cu. Because of this additional energy needed to precipitate, precipitation does not initiate when Si is just at the supersaturated point with Cu, but occurs only when it is supersaturated to a certain level [68,71].

Precipitation occurs when the temperature of the Si substrate is lowered, thus resulting in a decrease in the solubility of the impurities. The nucleation barrier for precipitation is reduced at specific sites such as crystal defects in the Si, including dislocations and oxide precipitates. As a result, precipitation occurs preferentially at these sites. Similarly, precipitation is possible in the device region and at the substrate surface due to the presence of crystalline lattice stress, surface defects, and at the interfaces of Si [72,73]. Precipitation of metals on the Si surface can result in haze formation [74]. In general, fast cool-down rates result in more and smaller precipitates than slow cool-down rates. This is a reversible process: if the temperature is raised, precipitation will redissolve into the Si [65].

The removal of metallic solutes from certain regions in the substrate is also known as "gettering." Apart from gettering by defects, impurities can also be gettered in regions of the device that have a higher solubility for the impurity than the bulk Si, such as highly doped regions and interfaces to polysilicon and Al. This is referred to as segregation gettering. Because segregation occurs below the solubility limit, it is possible to obtain lower concentrations by segregation gettering than by precipitation.

2.2.2.4.2 Metals in Dielectric Materials

The solubility and diffusivity of metals in SiO_x and SiN_x are much lower than those in Si [75,76]. Fig. 2.2-14 shows the diffusivity of various metals in SiO_2. Some metals interact with SiO_2 to form metal oxides or silicates, thus localizing the contaminant. This causes some dielectric layers to form effective diffusion barriers. However, the mobility can increase significantly if an electric field is applied. The metal is ionized and injected into the dielectric, where it drifts toward the cathode. This is a concern in Cu metallization [77], since Cu diffusion can result in a leakage path and eventually in dielectric breakdown [78]. This necessitates the use of a diffusion barrier; typical dielectric diffusion barriers are TiN and TaN. Mobile ions such as alkaline metals cause flatband shifts in gate oxides and other dielectrics. The mobilities

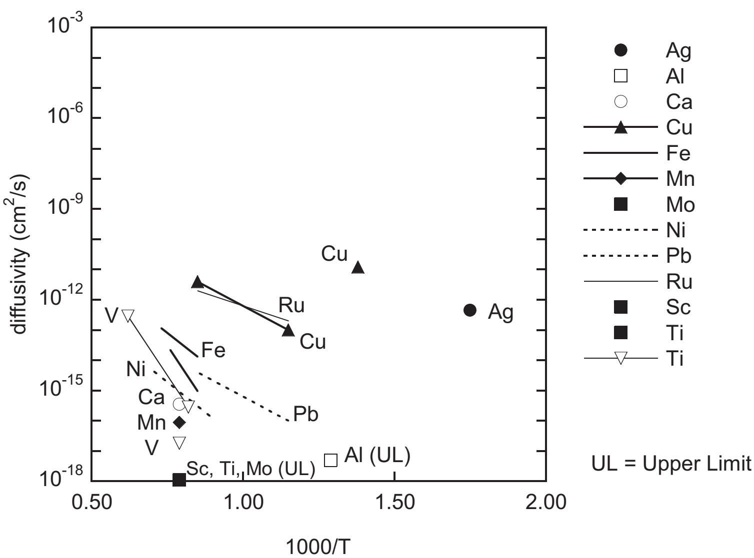

FIGURE 2.2-14 Diffusivities of various metals in SiO_2. *Compiled by the authors from data found in Refs: Al, Ca, Mn, Mo, Sc, and Ti [76], Ag and Cu [75], Cu, Pb, and Ru [96], Fe [83,218], Ni [219], and Ti [220].*

of these ions, especially Na^+ and K^+, are extremely high and detrimental to devices, as outlined in Section 2.2.2.5.

Low-k materials exhibit similar behavior to SiO_2, except their interaction with contaminants may be different. As a result, some metals may be present as ions in low-k materials, whereas they form silicates in SiO_2. For example, drift is observed for Al and Ta in an organosiloxane polymer, whereas no drift is observed in SiO_2 [79]. Since the metals can only drift when they are ionized, the drift rate depends on the ionic radius and on the ionization potential of the metal. This implies a dependence on the contamination source, because in some processes (e.g., ion implantation or plasma processing) the contaminant may be introduced in ionic form. The drift rate also depends on the low-k film type and quality. Generally speaking, dense materials with a high degree of cross-linking and polarity have lower drift rates [80].

Diffusion of metals through oxides is important for silicon-on-insulator (SOI) substrates. Most contaminants are likely to accumulate at the buried SiO_2 layer once they diffuse through the Si top layer [81,82]. The quality of the oxide layer may play an important role; damage sites and point defects may either enhance or retard the diffusion [83].

2.2.2.5 Effects of Metallic Contamination

Metallic contamination can have detrimental effects on device performance. This may be caused by incorporation of the contaminants into sensitive device regions, or by loss of process control induced by the contaminants.

2.2.2.5.1 Effect on Process Control

The presence of metallic contamination on a wafer surface can affect chemical reactions at the surface. Examples are changes in oxidation rates due to Al contamination [84,85], polysilicon deposition rates [86], and an increase of Si etch rates due to W or Ni contamination [87]. Room temperature oxidation of Si has been observed due to the presence of Cu_3Si precipitates [88,89]. Some phenomena caused by metallic contamination, such as enhanced oxidation, may cause device failures, even when the metallic contaminants would be relatively harmless. The impact of metallic impurities on gate oxides is often explained in terms of retarded oxide growth or oxide thinning [90,91]. However, there is limited experimental evidence to validate this hypothesis.

Metallic contaminants in SC-1 (described in Chapter 4) cleaning solutions can catalyze decomposition of the H_2O_2 in the mixture [92], which reduces bath lifetime and can enhance local etching of the silicon substrate [93,94], as shown in Fig. 2.2-15. The resulting increase in roughness affects the smallest size of particles that can be detected with light scattering tools and atomic force microscopy, as discussed in Chapters 12 and 13. For a more complete discussion on the effects of surface roughness, refer to Section 2.3.1.6.

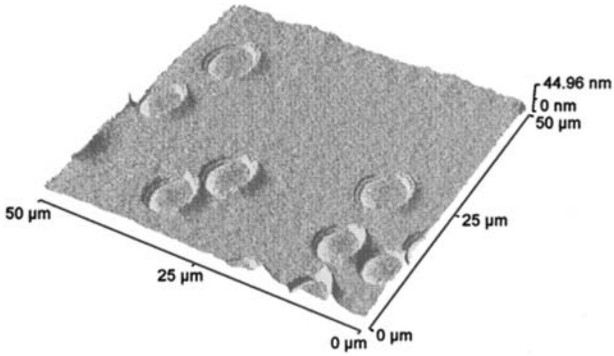

FIGURE 2.2-15 AFM plot of Si surface after immersion of a hydrophobic wafer into a 3 ppb Fe-contaminated SC-1 solution, 1:1:5 by volume [92]. *Reproduced by permission of ECS—The Electrochemical Society from P.W. Mertens, M. Baeyens, G. Moyaerts, H.F. Okorn-Schmidt, R. Vos, R. Waele, Z. Hatcher, W. Hub, S. De Gendt, M. Knotter, M. Meuris, M.M. Heyns, in: R.E. Novak, J. Ruzyllo (Eds.), Fifth International Symposium on Cleaning Technology in Semiconductor Device Manufacturing, vol. 97-35, Electrochemical Society, Pennington, New Jersey, 1997, p. 176.*

2.2.2.5.2 Bulk Silicon Defects

The presence of defects in bulk Si disturbs the periodicity of the lattice and thus introduces energy levels that are located within the bandgap, which can enhance the generation and recombination of minority carriers. Generation can be observed if the concentration of mobile charge carriers is low, a depletion condition. From Shockley-Read-Hall theory, the generation lifetime τ_{gen} can be written as [40,95]:

$$\tau_{\text{gen}} = \frac{2}{v_{\text{th}}\sigma_0 N_t}\cosh\left(\frac{E_t - E_i}{kT}\right) \tag{2.2-30}$$

where v_{th} is the thermal velocity, N_t is the volume density of defects, and $E_t - E_i$ represents the position of the defect energy level relative to the intrinsic Fermi level. In this equation, the hole and electron capture cross-sections of the defect are assumed to be equal to σ_0. T and k are the temperature and Boltzmann's constant, respectively. For a depletion region with width W the generation current density, J_{gen}, is

$$J_{\text{gen}} = \frac{qn_i W}{\tau_{\text{gen}}} \tag{2.2-31}$$

with q the electronic charge and n_i the intrinsic electron density. This current can be measured, for example, as a diode leakage current or as a transient of the depletion capacitance of a metal oxide semiconductor (MOS) capacitor.

Recombination can be observed if excess carriers are injected. With n and p the electron and hole concentrations, and assuming low injection conditions (concentration of minority charge carriers remains lower than the concentration of majority charge carriers), the so-called "Shockley–Read–Hall" recombination lifetime is given by

$$\tau_{\text{rec}} = \frac{1}{v_{\text{th}}\sigma_0 N_t}\left[1 + \left(\frac{2n_i}{n+p}\right)\cosh\left(\frac{E_t - E_i}{kT}\right)\right] \tag{2.2-32}$$

with n_i the intrinsic electron concentration and n and p the electron and hole density, respectively. The distance that a minority charge carrier L can diffuse before recombining depends on both the diffusivity D and the recombination lifetime τ_{rec}:

$$L = \sqrt{D\tau_{\text{rec}}} \tag{2.2-33}$$

The recombination lifetime affects the diffusion current in diodes, for example [96–98].

The expressions for generation and recombination lifetime are plotted in Fig. 2.2-16. The defect levels that are closest to the middle of the band gap are most effective in reducing the lifetime. The defects can be metal silicide precipitates, but also point defect complexes. The energy levels of many types of defects have been determined [99,100]. An example is the formation of Fe–B pairs in B-doped Si, which reduces the minority carrier lifetime. As a matter of fact, the Fe–B pairs can be dissociated by annealing at temperatures >200°C, resulting in a further reduction in lifetime. This difference in lifetimes has been calibrated and has become a standard method to determine the concentration of Fe in Si [101].

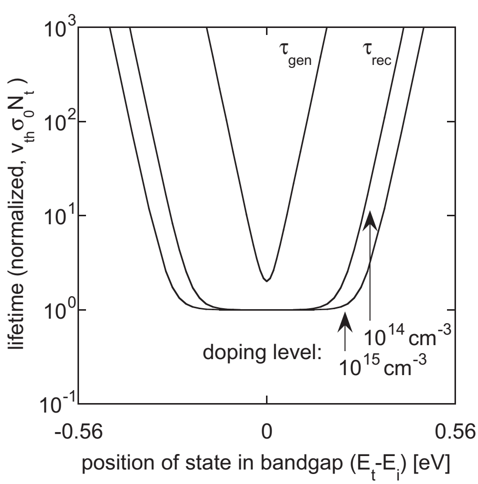

FIGURE 2.2-16 Normalized bulk generation and recombination lifetimes at room temperature in silicon, as a function of the position of the energy level within the band gap [40]. *S.M. Sze, Physics of Semiconductor Devices. Used with permission by the authors.*

2.2.2.5.3 Defects in Dielectrics

Gate dielectrics are characterized by their excellent insulating and capacitive properties. Metallic impurities on the wafer surface, prior to gate dielectric formation, usually degrade these properties by locally reducing the tunnel barrier or by introducing traps, thus forming a leakage path for charge carriers [62]. In other cases, impurities can locally retard oxidation rates, a phenomenon known as oxide thinning [90,91]. Similarly, a retardation of the gate dielectric growth rate could occur. If the impurities cause a significant surface morphology change, as may be the case for metallic particles, the electric field across the dielectric will be nonuniform [102]. Each of these mechanisms results in a local increase of the tunnel current density, which degrades the reliability of the dielectric.

When defects are present at the gate dielectric interface, charging of the defects affects the number of charge carriers in the channel of MOS transistors, resulting in "flicker noise" or carrier scattering. Generation and recombination velocities σ_{gen} and σ_{rec} can be defined in a manner analogous to Eqs. (2.2-29) and (2.2-31). However, surface generation and recombination is generally more complex than bulk generation and recombination [95]. Whereas bulk defects usually exhibit distinct energy levels within the band gap, surface defects introduce a broad distribution of interface state defect levels. Moreover, the generation/recombination activity of the surface defects is higher in depletion than in inversion or accumulation regions.

There are many publications discussing the effect of specific metallic contaminants on gate dielectrics. Among the most frequently investigated contaminants are Fe, Cu, and Ni. Iron especially is regarded a harmful contaminant, frequently being observed in production lines and causing device failures with concentrations as low as 1×10^{11}atoms/cm^2. The data for Cu are significantly more scattered, owing to the large diffusivity and hence the large dependence on processing conditions. Hence, the comparison of results found in different publications is not straightforward. In general it may be stated that metallic contaminants located near the gate dielectric will significantly affect the dielectric reliability. As a rule of thumb, harmful contaminants diffuse slowly in Si and/or form compounds at the surface such as oxides, silicates, or silicides. Examples are group II, alkaline earth, elements (e.g., Mg, Ca, Sr, and Ba), and light transition metals (Sc−Cu). Note that this rule does not always hold; although Al and Hf diffuse slowly and form metal oxides at the surface [84,103], they are generally not perceived as very harmful contaminants.

A special class of defects consists of ionized metallic impurities in dielectrics, which can drift under influence of an electric field. Best known are Na^+, K^+, and Li^+ contamination, usually of human origin or from packaging materials. In state-of-the art cleanroom facilities, these contaminants largely have been eliminated by elimination of their sources. However, Cu, used for interconnects, is also known to drift in oxides. Some of the consequences are variations in the surface potential, the threshold voltage, and local distortion of the electric field. The contamination can be detected by stressing the dielectric at fields of opposite polarity, followed by a capacitance measurement. The charge density ρ can be obtained from the shift in flatband voltage, V_{FB}:

$$\Delta V_{FB} = -\frac{1}{C_{ox}} \int_0^{t_{ox}} \frac{x}{t_{ox}} \rho(x) \mathrm{d}x \tag{2.2-34}$$

where x is the position in the oxide, t_{ox} is the oxide thickness, and C_{ox} is the oxide capacitance.

2.2.3 Atmospheric Molecular Contamination

Besides particles, there is a range of other species that may contaminate the wafer surface through the air. These species have molecular dimensions and are thus known as atmospheric molecular contamination (AMC) [104]. The main sources for AMC are outgassing of construction materials and tools, emissions from process chemicals, and make-up air contaminated by urban pollutants. Because of their molecular dimensions, AMC are difficult to filter out by conventional HEPA or ULPA filters; design of the airflow in the cleanroom is thus critically important in order to localize the contamination and prevent spreading it to sensitive areas. In addition, chemical filters consisting of active carbon or impregnated materials [105–107] can be used to remove organic contamination. Most of all, AMC has driven continuous improvements in construction materials.

In SEMI Standard F21-95 [107], different types of AMC are distinguished:

Acids: corrosive materials whose chemical reactions are those of electron acceptors.

Bases: corrosive materials whose chemical reactions are those of electron donors.

Condensables: chemical substances capable of condensation on a clean surface.

Dopants: chemical elements that modify the electrical properties of a semiconductor material.

It should be noted that this classification does not cover all atmospheric molecular contamination. For example, moisture is typically not considered as "condensable" and airborne metal contamination [108] does not fit the classification either. Moreover, some contaminants fit in more than one group, such as organophosphates that can be considered both condensables and dopants. The different types of AMC, in addition to moisture, are discussed in Sections 2.2.3.1–2.2.3.5.

2.2.3.1 Acids

The main sources for molecular acid contamination are chemical baths used for wet wafer processing. Typical examples are HF, HCl, HBr, HNO_3, H_2SO_4, and H_3PO_4. Sulfuric acid also can be formed from SO_2 in combination with moisture in the ambient atmosphere. Acids can deposit on the wafer surface by means of adsorption. It has been observed that HF and HCl have a strong affinity for deposition on Al and Cu layers [109]. Acids can cause corrosion of metal lines, which readily occurs at acid concentrations of a few ppb in the atmosphere. Therefore, a low level of molecular acid contamination is critical prior to metal sputtering and for all process steps with exposed metal layers. Also, trace levels of HF may end up in the air recirculation system and give rise to dopant contamination, as discussed in Section 2.2.3.4.

In the case of acid, base, and condensable contamination, it is often observed that deposition on surfaces causes hazing, especially in the presence of moisture [110–112]. This phenomenon is known as time-dependent haze. It affects particle detection on wafers, and also leads to nonuniform transmittance of light through the optical components in photolithography equipment and light scattering tools. This type of contamination on Si substrates is generally easy to rinse off in H_2O or evaporates during a heat treatment.

2.2.3.2 Bases

Airborne amines originating from processing mixtures containing chemicals such as hexamethyl disilazane, tetramethyl ammonium hydroxide, *N*-methyl-pyrrolidone, or ammonium hydroxide (NH_4OH) are the most common source of basic AMC. In addition, paints and adhesives may contain out-gassing solvents with alkaline properties. Low concentrations of amine contamination are critical for correct processing of chemically amplified photoresists. Since these resists rely on the photogeneration of acids, the presence of as little as 10 ppb of amines in the cleanroom air can cause its diffusion into the photoresist and neutralization of the photogenerated acids [113]. The result is the so-called "T-topping" or "skin-formation" of the photoresist, as shown in Fig. 2.2-17, leading to an increase in linewidth [115,116].

2.2.3.3 Condensables

The name condensable is mainly used for volatile organic chemicals. Apart from contamination from human sources, there are many other airborne organic contaminants. A recognized source is outgassing from construction materials, especially materials with a high porosity such as polytetrafluoroethylene (PTFE, also known as Teflon). Other dominant contaminants in cleanroom air are toluene, xylene, and aliphatic compounds [117]; however, these are normally not found on wafer surfaces. Instead, additives such as plasticizers, adhesives, and corrosion inhibitors [104,118,119] are often reported. Other sources for condensables are pump oil and cleaning solvents,

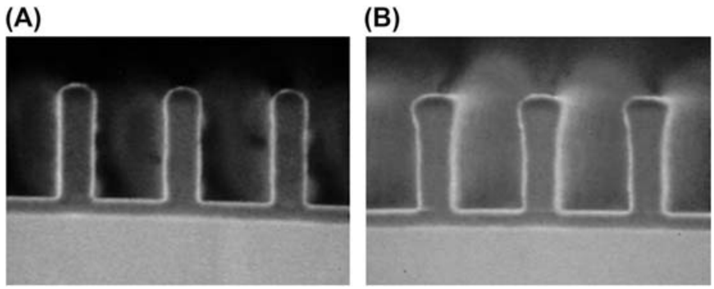

FIGURE 2.2-17 Impact of airborne amines in cleanroom air on 248-nm photoresist after exposure. Micrograph (A) shows a photoresist coating that was not exposed to amines; micrograph (B) shows a photoresist coating that was exposed to a contamination level of 15 ppb for 15 min [115]. *Used with permission from SPIE; D. Ruede, M. Ercken, T. Borgers, Molecular base sensitivity studies of various DUV resists used in semiconductor fabrication, Opt. Microlithogr. XIV, 4346 (1988).*

such as isopropyl alcohol (IPA). Prolonged exposures to a high vacuum ambient environment may cause deposition of vacuum silicone oil [120].

Photoresist is an obvious source for nonvolatile organic contamination. However, upon exposure to ultraviolet light volatile species may be formed [118,121–123] that can outgas in vacuum tools and that can deposit on optical components, thereby affecting the optical properties of the lithographic system. Incomplete evaporation or combustion of organic additives, such as trichloroethane, trichloroethylene, and dichloroethylene, during low-temperature oxidation can result in unwanted deposition of organic species on the wafer [124]. Examples are shown in Table 2.2-3, along with their applications and some properties [117,125–127].

Although the precise mechanism of adsorption is not understood, it is assumed that condensables with low boiling points (below 150°C) or with low molecular weights are of little importance because their residence time on the surface is relatively short [127]. Several other factors are known to play a role as well. A theoretical study has shown that the interaction between the Si surface and hydrocarbon molecule is expected to be stronger for larger molecules [128]. The adsorption and desorption characteristics of a number of organic contaminants has been studied [129,130]. There is a general tendency that polar molecules adhere more strongly to substrates than nonpolar molecules [110,117,123,131], especially if the surface also contains polar groups [132]. This may explain why different surface preparation techniques yield different amounts of adsorbed molecules [133].

Usually, organic contamination reacts in an oxidizing atmosphere or in aqueous or solvent cleaning mixtures, and therefore easily is removed. During high-temperature processes in oxidizing atmosphere, such as thermal oxidations, significant amounts of organics can be burned off [120,134]. However, detrimental effects, such as SiC formation at temperatures higher than 700°C,

TABLE 2.2-3 Examples of Common Airborne Molecular Contaminants and Their Effects on Wafers [134]

Category	Name		Chemical Formula	CAS No.	amu	Boiling Point (°C)	Vapor Pressure (Torr/25°C)	Application	Effects
Acids	Hydrochloric acid		HCl	7647-01-0	36	57	167	Process chemicals	Salt formation (time-dependent haze), corrosion on wafer and in HEPA filters
	Hydrofluoric acid		HF	7664-39-3	20	20	917		
	Nitric acid		HNO_3	7697-37-2	63	83	63		
Bases	Hexamethyl disilazane	HMDS	$C_6H_{19}NSi_2$	999-97-3	161	125	13.8	Process chemicals	T-topping, photolithography defects
	Ammonia		NH_3	7664-41-7	35	27	557		
	n-Methlypyrrolidone	NMP	C_5H_9NO	872-50-4	99	202	0.35		
	Tetramethylammonium hydroxide	TMAH	$(CH_3)_4NOH$	75-59-2	91	60–65	18		
Condensables	Dioctyl phthalate	DOP	$C_{24}H_{38}O_4$	117-81-7	391	387	1.4×10^{-7}	Plasticizers	Hydrophobie wafer surface, poor layer adhesion
	Diethyl hexyl phthalate	DEHP							
	Dibutyl phthalate	DBP	$C_{16}H_{22}O_4$	84-74-2	278	340	2.0×10^{-5}		

Continued

TABLE 2.2-3 Examples of Common Airborne Molecular Contaminants and Their Effects on Wafers [134]—cont'd

Category	Name		Chemical Formula	CAS No.	amu	Boiling Point (°C)	Vapor Pressure (Torr/25°C)	Application	Effects
	Butyl benzyl phthalate	BBP	$C_{19}H_{20}O_4$	85-68-7	312	370	8.3×10^{-6}		
	Diethyl phthalate	DEP	$C_{12}H_{14}O_4$	84-66-2	222	298	2.1×10^{-3}		
	Dioctyl adipate		$C_{22}H_{42}O_4$	103-23-1	371	417	8.5×10^{-7}		
	Isopropyl alcohol	IPA	C_3H_8O	71-23-8	60	97	21	Solvents	
	Toluene		C_7H_8	108-88-3	92	111	28	Paint	
	Xylene		C_8H_{10}	1330-20-7	106	138.5	8.0		
	Stearic acid		$C_{18}H_{36}O_2$	57-11-4	285	383	7.2×10^{-7}		
	Phenol		C_6H_6O	108-95-2	94	182	0.35		
Dopants	Tris(chloropropyl) phosphate	TCPP	$C_9H_{18}Cl_3O_4P$	13674-84-5	328	>270	2.0×10^{-9}	Flame retardants in	Counter doping, voltage shifts
	Trischloroethyl phosphate	TCEP	$C_6H_{12}Cl_3O_4P$	115-96-8	285	330	6.1×10^{-3}	HEPA filters	
	Boron silicate fibers	$B_xSi_yO_z$						HEPA filters	

Compiled by the authors from available data.

may occur if the temperature ramp-up is conducted in an inert environment, even with minimal amounts of *C* left on the wafer, or if the organic contamination is not completely burned off or vaporized during rapid thermal processing [110,131]. This can affect the gate oxide integrity [125,131,135,136]. Other detrimental effects may be observed in processes involving surface reactions: chemical vapor deposition, especially when performed in a reducing environment [118,119,135], such as variations in etching [110,131,137], wetting of the substrate resulting in an increase or decrease of the contact angle making the surface more hydrophilic or hydrophobic [110,111,118,119], and corrosion of metal layers [119].

2.2.3.4 Dopants

Contamination on the Si may occur due to inadvertent exposure to trace levels of impurities that are used to dope the Si to make the *p*- and *n*-regions of the wafer. For example, trace levels of HF may enter the air recirculation system and etch the borosilicate glass in HEPA filters [138]. The resulting BF_3 boron contamination that may fall onto the wafers can give rise to local variations in doping levels and thus fluctuations in threshold voltage. Similar doping defects can also be caused by outgassing of organophosphates from polyurethane sealants in the HEPA filters or from flame retardants [126,139,140].

2.2.3.5 Moisture

Relative humidity is defined as the ratio of the actual H_2O vapor pressure, p, over the saturation vapor pressure, p_s, at the prevailing temperature:

$$RH = \frac{p}{p_s} \tag{2.2-35}$$

where *RH* is the relative humidity. The saturation vapor pressure of water as a function of temperature is shown in Fig. 2.2-18. The relative humidity of cleanroom air is controlled close to 40%. This is required to keep electrostatic charging within acceptable limits [141]. The moisture adsorbs on the Si surface, which increases the surface conductivity and thus allows charge to dissipate through a natural conductance path to the ambient atmosphere.

Moisture adsorption is a material specific process and can occur at varying rates; for Si wafers the equilibrium is reached within minutes [133]. Borophosphosilicate glass [142], SiO_2, and high-k layers, such as ZrO_2 and HfO_2 [143], also adsorb moisture. The amount of H_2O molecules adsorbed on fully hydroxylated nonporous SiO_2 at 25°C and 40% RH is about 5 molecules/nm^2. This is approximately a monolayer, $\sim 5.6 \pm 0.2$ molecules/nm^2. Adsorption involves H-bonding to surface OH groups (silanols) but is physical and not chemical in nature, as it is completely reversible by heating the surface. If the SiO_2 is submitted to a thermal treatment the amount of H_2O molecules adsorbed is lower if the temperature of the treatment exceeds the range of

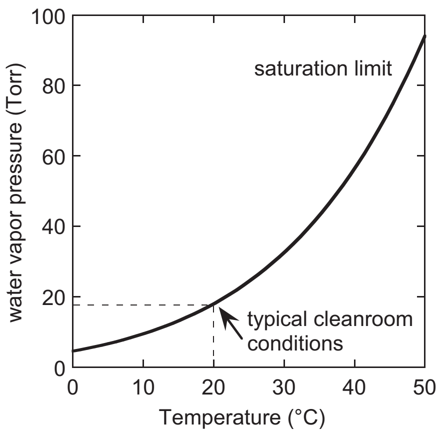

FIGURE 2.2-18 Water vapor pressure as a function of temperature [64]. In typical cleanroom conditions, a relative humidity of approx. 40% is used, which corresponds to a H_2O vapor pressure of ~20 Torr or 6000 ppm by mass. *Used with permission from CRC Press Inc.*

300–400°C. Partially dehydroxylated SiO_2 will slowly rehydroxylate when exposed to H_2O vapor.

Low-k dielectric materials are usually hydrophobic, and thus, not much moisture is adsorbed. However, after exposure to plasma, as in the case of photoresist stripping, polar silanol groups are formed at the surface, resulting in adsorption of H_2O, which increases the effective dielectric constant of the layer. For this reason repair of the low-k layer is done by removing the thin damaged layer [144]. Alternatively, the layer can be sealed; the pores are closed at the surface to reduce the exposed area where H_2O can adsorb. Chapter 8 discusses the process of repairing and sealing the low-k material.

2.3 SOURCES OF DEFECTS AND CONTAMINATION

In the following sections, we will discuss several media and process tools that can be sources of contaminants and defects.

2.3.1 Cleaning-Related Defects

2.3.1.1 Megasonic Damage

Megasonic energy is used to enhance the removal of contaminants, primarily particulate contamination. It is a physical force, and as such it is nonselective with regard to device structures; hence not only contaminants but also device structures may be removed or damaged when this method is applied. The mechanism underlying megasonic-induced damage is not completely understood, but it is often assumed that a damage threshold for a specific energy density exists. This threshold would depend, among others, on the pitch and dimensions of the device structures [145]. Chapter 3 discusses the causes of megasonic damage with respect to the particle removal process. Chapter 4 discusses the megasonic systems and processes. A systematic study is complicated because the megasonic energy density is not uniform, but exhibits a pattern that is strongly dependent on the piezoelectric transducer configuration; Chapter 4 discusses various arrangements of these transducers. Also, it has been argued that damage primarily occurs on delicate structures and at weak spots in the device features, such as polySi grain boundaries or areas with poor layer adhesion, which implies a dependence on process control [146].

2.3.1.2 Transient Effects During Rinsing

When a substrate is transferred from a cleaning mixture to a rinse bath, it carries over a chemical layer that contains cleaning agents and possibly contaminants. When the wafer is immersed in pH-neutral water, the carry-over layer changes from acid or alkaline pH to neutral pH. This may cause contamination in the carry-over layer to redeposit on the substrate surface [147]. In the case of rinsing after amine-based cleaning, the pH value may increase causing corrosion of metal layers. Assuming only diffusive transport and a static rinse tank, the concentration of either H_3O^+ ions or contaminants at the surface $C(t)$ can be approximated by [34]

$$\frac{C(t) - C_{\text{rinse}}}{C_0 - C_{\text{rinse}}} \approx \frac{\delta}{\sqrt{\pi D t}} \quad \text{with} \quad \frac{\delta}{\sqrt{\pi D t}} \ll 1 \tag{2.3-36}$$

where C_{rinse} is the concentration in the rinse bath and C_0 the initial concentration in the carry-over layer. The thickness δ of the carry-over layer is typically 20 μm. From the Stokes-Einstein equation, the diffusivity, D, of H_3O^+ in water is $D = 1 \times 10^{-4}$ cm²/s; in contrast, for a particle of 0.1 μm diameter $D = 4 \times 10^{-8}$ cm²/s and for most metal ions $D = 1 \times 10^{-5}$ cm²/s. Thus, the pH value will initially change very quickly, before the contaminants diffuse away from the surface; afterwards the change will proceed more slowly. The change in pH may therefore induce redeposition of contaminants onto the wafer surface.

2.3.1.3 Drying Residues

In the last stage of the drying process, H_2O evaporates and leaves behind any nonvolatile residues that were present in the evaporating H_2O layer. In the case of spin-drying, the thickness of this water layer is of the order of 5 μm, whereas for Marangoni drying it is two orders of magnitude lower [148]. Chapter 4 outlines these different drying techniques. The residues may be metallic or particle contamination, or cleaning agents that were not completely removed. One type of residue is known as "watermarks," which originates from silica that is formed and dissolved during the rinse cycle [149]:

$$\begin{aligned} & Si + O_2 \rightarrow SiO_2 \\ & SiO_2 + H_2O \rightarrow H_2SiO_3 \end{aligned} \tag{2.3-37}$$

Watermarks usually occur in hydrophobic regions and become more pronounced if a hydrophobic/hydrophilic pattern is present on the substrate. Possible effects include local variations in oxide thickness, adhesion problems, and increased contact resistance [150]. When small amounts of HF are present, either from the ambient atmosphere or from insufficient rinsing, the watermarks have a more crystalline morphology due to the formation of H_2SiF_6. It has been shown that watermarks can be suppressed by using N_2-saturated rinse water at low pH [151].

2.3.1.4 Pattern Collapse

In the last stage of the drying process, small amounts of H_2O can give rise to capillary forces between structures. The magnitude of these forces depends on the distance between the structures and the surface tension of the medium. For high-aspect ratio structures they can be large enough to cause pattern collapse, a deformation of the structures that usually results in failure. The equivalent capillary force F_{cap} that is exerted on the upper edge of the line can be written as [152]

$$\frac{F_{cap}}{l} = \frac{\gamma \cos \theta}{s} \tag{2.3-38}$$

where θ is the contact angle, γ is the liquid surface tension, and l and s are the length and spacing of the lines, respectively. If the capillary forces on both sides of the lines are in equilibrium, and if the spacing varies, or if liquid removal is nonuniform, a resultant force exists, which causes a deformation δ of the line:

$$\delta = \frac{4}{E} \frac{F_{cap}}{l} \left(\frac{h}{w} \right)^3 \tag{2.3-39}$$

where E is the Young's modulus, h is the height of the line, and w is the width of the lines. Eq. (2.3-38) is only valid for small deformation, as it does not take into account that the deformation reduces both the spacing s and the contact angle θ, with a further increase of F_{cap}. If the forces become large enough, the plastic yield strength and the adhesion of the lines may become important [152]. Fig. 2.3-19 shows an example of pattern collapse of dense structures. To avoid pattern collapse, surfactants such as IPA or other rinsing agent may be used to decrease the surface tension. For the same reason, the use of supercritical CO_2 has been investigated as a replacement for H_2O in certain process steps [153].

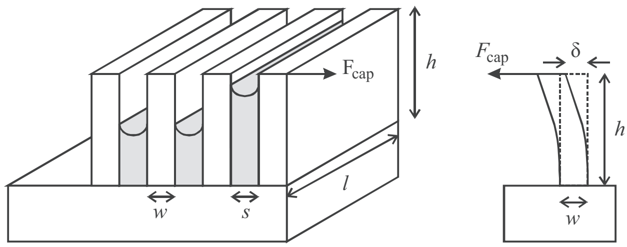

FIGURE 2.3-19 Schematic view of line structures and capillary forces. *Used with permission from authors.*

2.3.1.5 Substrate Charging

A particular concern with drying is triboelectric charging of the substrate [154], because substrate charging enhances particle deposition and can also initiate a fast charge transfer (electrostatic discharge), resulting in excessive junction leakage, oxide breakdown, or metal fusing [141]. If different regions of the substrate are at different potentials, corrosion can occur. Triboelectric charging is an issue with spin-drying and brush cleaning, and it is also known to occur during Marangoni drying. Additionally, tool malfunctioning due to substrate charging has been reported as a problem [155].

Substrate charging can be controlled in a number of ways. Air ionizers are able to neutralize substrate charge, but there have been concerns about their contribution to particle contamination levels. Humidity provides a conducting path to ground by forming a layer of adsorbed moisture on surfaces; however, it also increases the risk of corrosion or of forming time-dependent haze. The use of antistatic and dissipative materials (e.g., polycarbonate) in tools has a similar effect, but again the particle performance of these materials has been debated. Guidelines for acceptable static charge levels are available in the SEMI E78 standard [156].

2.3.1.6 Surface Roughness

Because cleaning can involve etching of the substrate, it usually causes a slight increase in surface roughness. The roughness can become more pronounced if impurities are present either in the cleaning mixture or on the substrate itself. An example is the decomposition of H_2O_2 in metal-contaminated SC-1 cleaning mixtures; the decomposition results in evolution of H_2 gas bubbles that adhere to the wafer surface and cause inhomogeneous etching of the substrate [93], as shown in Fig. 2.2-17. Another example is the local dissolution of Si in Cu-contaminated HF mixtures, as discussed in Section 2.2.2.3. Surface roughness is most easily detected as an increase in the haze level using light scattering tools. A direct consequence of this reaction is an increase in the minimum size of particles that can be detected.

The impact of roughness on devices continues to be a concern. However, it should be noted that many studies on the topic artificially increase the roughness to above 1 nm RMS (root mean squared) values, which is large compared to what is typically required in production lines [20]. It is known that the tunnel current through gate dielectrics increases due to roughness. Usually this is attributed to distortion of the electrical field [157,158], but an alternative explanation in terms of oxide thickness variations also has been proposed [159].

The effect of field distortion is expected to be largest in the Fowler–Nordheim tunneling regime, where it induces variations in the tunnel distance. In contrast, in the direct tunneling regime the tunnel distance is constant, and field distortion has a less pronounced effect. For this reason, and because thermal oxidation has a slightly planarizing effect on roughness, roughness may be expected to have a larger impact on deposited layers (e.g., high-k) than on thermally grown ultrathin oxide layers [160]. It also has been noted that roughness affects the surface mobility of charge carriers [161]. Under strong inversion and high fields the mobility is limited by scattering due to roughness at the surface. It is found that, on Si <100> substrates, the roughness-limited mobility $\mu \propto N_s^{-2}$, where N_s is the carrier density in the inversion layer [162,163]. Scattering at the buried oxide interface also occurs in SOI devices, which becomes significantly more pronounced if the Si thickness decreases [164,165].

2.3.1.7 Corrosion

Corrosion of metal lines is a widespread problem for microelectronic device fabrication [16]. It is an electrochemical reaction, and thus involves both oxidation and reduction reactions. Corrosion proceeds due to a potential that exists between two metal regions; this may be a galvanic potential or it may be due to charging of the regions. Charging occurs during operation of the device, but it may also occur during device processing, for example after plasma treatment or stripping [166–168].

The dynamics of electrochemical corrosion are described by the Nernst equation

$$\Delta G = \Delta G^0 - RT \ln Q \tag{2.3-40}$$

where G is the Gibbs free energy, R the universal gas constant, and T the absolute temperature. Q is the reaction coefficient for the general reaction:

$$aA + bB \rightarrow cC + dD$$
$$Q = \frac{a_C^c a_D^d}{a_A^a a_B^b} \tag{2.3-41}$$

The constants a_A^a, a_B^b, a_C^c, and a_D^d are the activities of the respective species raised to some species-specific power. For solids and liquids, the activities are taken as unity. For gases the activities are usually taken as the partial pressure, and for dissolved species as the molar concentration. The free energy change ΔG is related to the electrical oxidation potential U by Faraday's law:

$$\Delta G = -nFU \tag{2.3-42}$$

where F is the Faraday constant and n is the number of electrons participating in the reaction. The corrosion of metals is determined by the oxidation potential, as shown in Table 2.3-4, but is also promoted by the presence of mechanical stress or grain boundaries.

Possible oxidation reactions at the anode are

$$\begin{aligned} &\mathrm{Mt} \rightarrow \mathrm{Mt}^{n+} + n\mathrm{e}^- \\ &\mathrm{Mt} + n\mathrm{OH}^- \rightarrow \mathrm{Mt(OH)}_n + n\mathrm{e}^- \\ &\mathrm{Mt} + n\mathrm{H_2O} \rightarrow \mathrm{MtO}_n + n\mathrm{H}^+ + n\mathrm{e}^- \end{aligned} \tag{2.3-43}$$

Depending on the conditions the metal, Mt, can either form an oxide layer or dissolve. The formation of a metal oxide layer, as in the case of Al, Ni, Ti, and steel (Ni, Cr, and Fe), can protect the metal from further corrosion, but at the same time it increases the contact resistance. Dissolution of the metal line obviously is undesirable because ultimately it can lead to open circuits.

The accompanying reduction reaction at the cathode can be a reduction of dissolved metal cations, reduction of adsorbed H_2O molecules, or formation of metal hydroxides:

$$\begin{aligned} &2\mathrm{H}^+ + 2\mathrm{e}^- \rightarrow \mathrm{H_2} \\ &\mathrm{O_2} + 4\mathrm{H}^+ + 4\mathrm{e}^- \rightarrow 2\mathrm{H_2O} \\ &\mathrm{O_2} + 2\mathrm{H_2O} + 4\mathrm{e}^- \rightarrow 4\mathrm{OH}^- \\ &\mathrm{Mt}^{n+} + n\mathrm{e}^- \rightarrow \mathrm{Mt} \\ &\mathrm{Mt} + (n+x)\mathrm{H_2O} + x\mathrm{e}^- \rightarrow \mathrm{Mt(OH)}_{n+x}^{-x} + \frac{x}{2}\mathrm{H_2(g)}n\mathrm{H}^+ \end{aligned} \tag{2.3-44}$$

Metal deposition is undesirable because it can cause short circuits between metal lines, whereas formation of soluble metal hydroxides can cause open circuits.

The charge transport required for these reactions occurs in the H_2O layer that is adsorbed on the surface. Moisture, therefore, has a strong impact on the corrosion rate. It can originate in the cleanroom environment, but it can also penetrate the device package during operation. The pH of the H_2O layer is reduced near the anode and increased near the cathode. As the dissolved metals migrate toward the cathode, the pH increases and metal precipitates may form. It should be noted that at low pH or in absence of O_2 (in a reducing atmosphere) the passivating metal oxide film might be unstable, thus increasing the dissolution rate of the metal. Especially for Cu, the dissolution

TABLE 2.3-4 Oxidation/Reduction Potentials of Metals [64]

Reaction		Product	Electrode Potential (eV)
$O_2 + 4H^+ + 4e^-$	→	$2H_2O$	1.229
$O_2 + 2H_2O + 4e^-$	→	4OH	0.401
$O_2 + 2H_2O + 2e^-$	→	$H_2O_2 + 2OH^-$	−0.146
$2H_2O + 2e^-$	→	$H_2 + 2OH^-$	−0.8277
$2H^+ + 2e^-$	→	H_2	0
$Au^+ + e^-$	→	Au	1.692
$Au^{3+} + 3e^-$	→	Au	1.498
$Cl_2(g) + 2e^-$	→	$2Cl^-$	1.35827
$Pt^{2+} + 2e^-$	→	Pt	1.18
$Pd^{2+} + 2e^-$	→	Pd	0.951
$Ag^+ + e^-$	→	Ag	0.7996
$Cu^+ + e^-$	→	Cu	0.521
$Cu^{2+} + 2e^-$	→	Cu	0.3419
$Fe^{3+} + 3e^-$	→	Fe	−0.037
$Pb^{2+} + 2e^-$	→	Pb	−0.1262
$Sn^{2+} + 2e^-$	→	Sn	−0.1375
$Ni^{2+} + 2e^-$	→	Ni	−0.257
$Co^{2+} + 2e^-$	→	Co	−0.28
$Cd^{2+} + 2e^-$	→	Cd	−0.403
$Fe^{2+} + 2e^-$	→	Fe	−0.447
$Cr^{3+} + 3e^-$	→	Cr	−0.744
$Zn^{2+} + 2e^-$	→	Zn	−0.7618
$Cr^{2+} + 2e^-$	→	Cr	−0.913
$Mn^{2+} + 2e^-$	→	Mn	−1.185
$Ti^{3+} + 3e^-$	→	Ti	−1.37
$Ti^{2+} + 2e^-$	→	Ti	−1.63
$Al^{3+} + 3e^-$	→	Al	−1.662

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at pH 2−5 is very rapid and the formation of a passivating oxide layer is hindered [168,169].

Contamination from the acid and base chemicals should also be considered. Species adsorbed on the metal, such as NO_x, SO_2, HF, Cl_2, HCl, H_2S, and amines, have an effect on pH and thus affect corrosion rates. Also, besides attracting moisture from the ambient environment (hygroscopic action), these species increase the conductivity of the adsorbed H_2O layer, and thus accelerate corrosion [170]. Moreover, many of these contaminants can attack metal oxides to form soluble metal salts [168].

2.3.2 Process Gases

In general, process gases used for microelectronic manufacturing can be manufactured with high purity (99.9999%). The most common impurities in gases are O_2, CO_2, and, especially for specialty gases, reaction by-products. The distribution of process gases from the manufacturing facility to the point-of-use (POU) brings into play several mechanisms that can load the gases with more contaminants. These mechanisms are related to the design and implementation of the distribution system. The main causes for additional process gas contamination are leakage from the environment and insufficient purging after system maintenance. Material outgassing can also be considered a contributor; however, baking of a new system greatly reduces this factor.

The main concern with process gases is moisture. Trace amounts can accumulate on the inner surface of the tubing. This is the case if the surface is rough, if dead space is present that cannot be adequately purged, or if improper welding results in stress or cracks in the surface [171]. Obviously, the choice of material is important for reducing moisture accumulation. Normally, electropolished steel with a smooth surface is used. Similarly, steel with a high Mn content is best avoided [172]; the high vapor pressure of Mn causes its evaporation and redeposition during welding of the tubing. The redeposited Mn increases the surface roughness and enhances adsorption of moisture. Moisture adsorption can also be induced if the gas experiences a transition from high to low pressures (Joule−Thomson effect) [162]. Moisture accelerates corrosion, especially if the gas is corrosive such as HCl (see Section 2.3.1.7). Metallic corrosion products can contaminate the gas flow as both particles and volatiles (e.g., MoCl, FeCl, and TiCl), depending on their relative vapor pressure [162].

Reaction by-products in specialty gas, such as those used for etching or deposition, may have a different vapor pressure than the specialty gas itself [173]. The result is that the impurity fraction delivered by the system varies with time. Moreover, process gases are not the only possible origin of impurities in the process environment. Impurities such as moisture and O_2 can also originate from the cleanroom air during wafer loading and from any exposed surface in the process chamber, including the wafer itself [174]. These impurities can directly affect process control rather than induce localized defects. Pumping down a vacuum chamber to low enough baseline pressures helps to limit these impurities.

2.3.3 Process Liquids and Photoresist

As with process gases, a distinction can be made between the purity of incoming chemicals at the "point of distribution" and the purity of the chemicals at the POU. Currently, most commonly used aqueous-based chemicals (NH_4OH, HCl, HF, H_2O_2, and H_2SO_4) are available with specifications <100 ppt for cations. For ultrapure water (UPW) the levels are 1−2 orders of magnitude lower, whereas in organic solvents and photoresist, the levels are often 2−3 orders of magnitude higher.

The contaminant concentration in liquid chemicals can increase due to leaching of cations and hydrocarbons (plasticizers and additives) from construction materials. PFA (perfluoroalkoxy) and PTFE are preferred materials for distribution systems and filters, whereas wet benches and tool parts are often made of polyvinyl chloride and polypropylene (PP). Not only are these materials compatible with a wide range of chemicals, but they also have a high density, low porosity, and therefore low contact area with the liquid. Typical leach rates for metals are 1×10^{12} atoms/day per cm^2 area of material [175,176], while more aggressive chemicals (concentrated HF, H_2SO_4/H_2O_2 mixtures) have higher leach rates. Because the leach rate decreases over time, materials are usually preconditioned before use. Quartz components contain <1 ppm of most metallic contaminants, with somewhat higher concentrations of Al. Cleaning solutions such as SC-1 may etch the quartz at rates of ~0.1 nm/min [177] thereby releasing $\sim 1 \times 10^{11}$ atoms/day per cm^2 of quartz or less than 1 ppt into the cleaning solution.

Another detrimental contaminant in UPW is dissolved O_2 gas. Sources for O_2 contamination can be gaseous permeation of the distribution system, or exposure to air in the process tool. Use of a N_2 blanket can prevent this. Dissolved O_2 causes oxidation of hydrogenated surfaces, which affects layer deposition. The contamination of UPW by silica should also be noted, the contamination can be caused by dissolution of oxide layers on the wafer. As outlined in Section 2.2.3.1, dissolved silica can cause drying residues.

Little is known about the interaction of contaminants with Si substrates in nonaqueous chemistries. It has been suggested that metals can form complexes in photoresist [178]. Furthermore, adsorption of metals on Si surfaces occurs in a similar manner as in aqueous solutions [178]. More recently, it has been found that the contamination level in commonly used solvents is dominated by the evaporation rate of the solvent [179]. With supercritical CO_2 under consideration as a medium for certain applications, a number of studies have been carried out on the fundamental interactions of contaminants with various organic solvents. Chapter 8 discusses the use of supercritical fluids for surface conditioning.

2.3.4 Ion Implantation

When a material is exposed to energetic species, as is the case with ion implantation, small amounts of the material may be sputtered off and redeposited elsewhere. Apart from the processed wafer, the exposed material can be any tool part. Typical contaminants are Fe, Cr, Al, Na, Cu, Mo, and W, often as particles [180]. Because the sputtered material becomes energetic, it may be implanted several tens of nm into the substrate, such as ion implanters could accomplish [181−183]. This means that subsequent cleaning does not always help in removing the contamination. Instead, a screening oxide is usually deposited, which is etched away along with the contamination.

There are several mechanisms that participate in the contamination process:

Contaminants are sputtered off, ionized, and accelerated towards the wafer. This is the case if the residence time of the contaminants is long enough for ionization. In practice this does not occur often and the contaminants adsorb elsewhere in the system.

Direct line of sight sputtering onto the wafer occurs when there is a straight path from the sputtered contaminant to the wafer. The most important contributors to this mechanism are therefore tool parts such as clamps, Faraday cups, and chamber walls. Typical contaminants are Al and stainless steel components such as Fe, Ni, and Cr [180,184−187]. The contamination level increases with implantation dose and also depends on the sputtering rate by the primary implanted species.

Multi-step sputtering involves the repeated sputtering and redeposition of contamination within the system until it reaches the wafer surface. This actually is an equilibrium between sputtering and redeposition rates; an equilibration effect may be expected if the plasma chemistry or the implanted species is changed [187]. Like the direct sputtering mechanism, this mechanism induces tool-related contamination. However, it is also responsible for cross-contamination between wafers. Examples are the sputtering of Co and Cu during contact and via etching [78,188] or the memory effect after changing the implantation source [183].

Over years, the contamination due to sputtering has been substantially reduced by application of low-contamination liner materials such as Teflon [181], quartz [185], silicon [184,187,189], and graphite [182]. However, the wafer itself is still a source of contamination. Also, contaminants that originate from the ion source and have the same mass/charge ratio as the implanted ion will not be screened by the analyzing magnet. Table 2.3-5 shows a number of interfering ions. In the case of plasma doping, the analyzing magnet is absent and all ionized impurities will be coimplanted. For this reason it is preferred to dedicate one process chamber for each implanted species [182].

TABLE 2.3-5 Overview of Common Co-implantation Species and Their Sources

Primary Beam	Contaminant	Source	References
BF_2^+	$^{98}Mo^{2+}$	Sputter source	[221,222]
	W^{2+}	Chamber walls	[221]
	$^{11}B^+$	Collisions and dissociation of BF_2^+	[223]
^{28}Si	$^{+14}N_2^+$	Air leak	
$^{31}P^+$	$^{62}Ni^{2+}$		

2.3.5 Plasma Reactors

The comments that were made for ion implantation regarding sputtering of contaminants are equally valid for plasma processing (etching, deposition, or resist stripping), except that the contaminants are not implanted as deeply into the substrate [190−192]. However, there are several other types of contamination associated with plasma processing.

2.3.5.1 Particle Contamination

Particles may be formed inside the reaction chamber due to flaking of deposited material from the chamber walls, material sputtered, or reaction of process gases with any material in the chamber. The smallest particles are formed from the process gases at the substrate surface. After formation, they travel away from the surface, and grow by agglomeration [193−195]. Finally, they become trapped in the plasma sheath. The motion of the particles is governed by a combination of thermophoresis (i.e., a temperature gradient) and electrostatic interaction [196−198]. It should be noted that most particles are charged, which will affect the way they interact with the plasma. Once the plasma is turned off, the particles can fall down onto the substrate if they are sufficiently massive [199,200]. Particles formed in plasma deposition tools can increase surface roughness.

2.3.5.2 Post-etch Residue and Resist

During reactive ion etching using perfluoro compounds (PFC), polymeric residues are formed that deposit on the substrate and chamber parts during etching. Sometimes, it is a desirable feature of the technology that these polymers also deposit on device structure sidewalls and protect the structure from lateral etching. However, when not removed, the residues can cause an increase of interconnect resistance, current leakage [201], and other undesirable electrical problems.

The post-etch residues typically consist of a combination of C, F, O, Si, and any other material that is exposed to the plasma during etching [202,203] that may be sputtered onto the wafer surface. The inclusion of metallic or other nonvolatile species makes removal of the residues more difficult. Chapter 7 outlines the post-etch polymers that may be encountered. Plasma stripping can remove some of the residues, although for complete removal of the residues an additional liquid cleaning step involving solvents or aqueous-based solutions is usually required. Each of these methods has several negative effects, making residue removal very complex [204]. Chapter 7 describes the plasma stripping and cleaning processes and Chapter 4 deals with wet cleaning and stripping techniques.

Oxygen plasma not only removes photoresist and residues, but also oxidizes the substrate. This may increase the contact resistance of conductor layers, thus making an additional cleaning step necessary. Resist stripping with O_2 plasma is successfully performed on many films without detrimental effects. However, exposure to O_2 plasma reduces the hydrophobicity of organic low-k dielectrics, such as carbon-doped oxide (CDO). Porous low-k dielectrics—rendered hydrophilic will then take up moisture, which is difficult to remove, resulting in an increase of the dielectric constant. Additionally, *C* is abstracted from the CDO material, causing a further increase of the dielectric constant. Treatment may be required to repair the damaged low-k layer [143,203].

Alternatives to oxygen plasma exist, such as argon sputter cleans, NF_3, or reducing chemistries ($He-H_2$, N_2-H_2, and NH_3) [202,205] (see Chapter 6). However, such chemistries may also attack chamber parts and cause particle contamination [206] or damage the underlying low-k layer.

The wet cleaning mixtures that are used after oxygen plasma stripping often contain amines and organic solvents. Without countermeasures, the amines can cause severe corrosion of metal layers [119,166−168,201]. Corrosion can reduce yield due to an increase of contact resistance, or undercutting of dielectric layers. Therefore, corrosion inhibitors are frequently added. Sulfuric acid/hydrogen peroxide mixture (SPM) is used to remove the remaining organic residue after plasma stripping, typically when metals are not present. A specific problem with SPM is its high viscosity, making it difficult to rinse from the wafer surface after the cleaning step. This then may give rise to time-dependent haze (see Section 2.3.1). The use of SPM is further discussed in Chapter 4.

In addition, the entire process from etching with PFC chemistry to solvent-based liquid cleaning poses an environmental waste problem and a risk for safety and health [207]. Alternative etching formulations [208] and aqueous cleaning mixtures [168,203] are therefore desirable. Also the use of supercritical CO_2 for residue removal is investigated, refer to Chapter 8. The large range of formulations, with different physical and chemical properties, requires a careful choice of tool materials and filters to ensure optimum resist stripping while avoiding degradation or contamination issues [209].

A particular difficulty with the evaluation of residue removal cleaning efficiency is evaluating the cleanliness of contacts, vias, and deep trenches. For representative results, cross-section scanning electron microscopy (SEM) conducted at different positions on the wafer may be required [203]. An additional complication is that residues can react over time with ambient air, giving rise to uncontrolled variations in cleaning performance. Also, thickness variations of photoresist and density variations of vias can impact the amount of residues in the features [19].

2.3.6 Wafer Handling, Transport, and Carriers

In many cases physical contact between the wafer and a tool is required, either to transport the wafer from one place to another (using vacuum tweezers or end effectors) or to hold the wafer in a fixed place (using wafer carriers, chucks, etc.). If a wafer is contaminated, it may transfer part of the contamination to the tool, which can then spread the contamination to other wafers. For this reason, it is desirable to minimize the contact area between the wafer and the tool. Important factors are the contact force, the material properties, and the surface morphology of the tool and the wafer.

Physical contact is also a direct source of contamination caused by abrasive or adhesive wear. The area of contact between two solid surfaces is limited to points of contact between surface asperities. The load applied to the surfaces will be transferred through these points of contact and the localized forces can be very large. This results in plastic flow of the softer material (either tool or wafer), which generates particles and scratches. The total volume V_{wear} of material that is generated by wear can be approximated by

$$V_{\text{wear}} = K \frac{F_{\text{load}} L}{H} \tag{2.3-45}$$

where F_{load} is the normal force at the interface, L is the sliding distance, H is the hardness of the material, and K is a wear coefficient that depends on material properties, temperature, and the presence of lubricants. For example, it has been found that particulate contamination from polyetheretherketone is significantly less than that of PP or polybutylene terephthalate [43,210].

2.3.7 Thermal Processing

In the past, contaminants originating from, heater elements for example, were known to diffuse through the quartz tube and deposit on the wafer surface [211]. Currently, through careful design and selection of materials, furnaces are usually very clean. Instead, processes can induce contamination, such as flaking of deposited material from the furnace walls or gas-phase particle formation. Also, contamination on wafers can evaporate at elevated temperatures and redeposit elsewhere in the furnace. Using the vapor pressure of the contaminant [64] and the ideal gas law, estimation of the maximum equilibrium metal concentration in the gaseous phase can be performed. If all gas-phase contaminants in a 1-cm thick layer above the wafer surface would deposit on the wafer, then a calculation of the surface concentration on the wafer, yields, according to this rough estimate, a vapor pressure of approximately 1×10^{-4} Pa, which results in approximately 1×10^{10} atoms/cm^2 on the wafer surface. At a temperature of 1000°C, many species have a lower vapor pressure. Notable exceptions are group I and group II metals (Na, K, Ca, Mg, Sr, and Ba), and Al, Zn, Pb, and Mn. Some metal oxides and also metal chlorides have higher vapor pressures (however, addition of Cl during dry oxidation is actually used to remove residual contaminants from the wafer).

2.3.8 Vacuum Processing

Vacuum systems are essential for plasma processing and thin layer deposition. They allow a precise control over processing conditions and eliminate trace amounts of contamination such as particles, moisture, and unwanted gases. The use of ultra-high vacuum wafer carriers actually has been proposed for better control over wafer contamination [212]. An overview of various vacuum systems in semiconductor manufacturing is given in Table 2.3-6.

At the same time, vacuum systems operating under high and ultra-high vacuum conditions are very sensitive to contamination in the vacuum chamber. In most cases, the contamination originates from outgassing of construction materials (e.g., H_2 gas, carbon monoxide, hydrocarbons), from permeation of gases through seals (e.g., N_2, O_2, and H_2O), or from desorption of molecules that are adsorbed on the wafer or the chamber walls (e.g., H_2, moisture, solvents). Moisture can originate from venting with gas containing moisture, but also from the wafer surface itself.

The most obvious effect of these contaminants is to increase the pump down time, although contaminants may also adsorb on surfaces. When performing SEM analysis, these contaminants can reduce the resolution. In sputter systems, contaminants can adsorb on the target while the system is idle, so that the first few nanometers of a sputtered layer is relatively rich in contaminants [213]. Hydrocarbons deposited on wafers can affect subsequent processing steps [194]. Adsorbed contaminants, such as AMC, can cause corrosion of metallic components of the vacuum system or of metal layers on the wafer itself.

Adsorption can be dominated by physical (van der Waals) forces or by chemical forces. In contrast to chemical adsorption, physical adsorption is always reversible and not limited to adsorption of one monolayer. The rate of

TABLE 2.3-6 Overview of Vacuum Applications in Semiconductor Manufacturing

Process Type	Pressure Range (Torr)	Pressure Range (Pa)	Time Required for Monolayer Adsorption (s)
Wet processes	579	10^5	
Oxidation			
High-temperature epitaxy			
Lithography	$1–10^{-3}$	$10^2–10^{-1}$	
Chemical vapor	$10^{-3}–10^{-5}$	$10^{-1}–10^{-3}$	$10^{-3}–10^{-1}$
Deposition			
Low-temperature epi			
Plasma etch			
Metallization (sputtering)			
Ion implantation	$10^{-6}–10^{-8}$	$10^{-4}–10^{-6}$	$10^0–10^2$
Metrology (SEM)	$<10^{-9}$	$<10^{-7}$	$10^3–10^5$
Molecular beam epitaxy			

adsorption of molecules at a surface, R_c, is proportional to the flux of incident molecules J_c and to the condensation coefficient α_c:

$$R_c = a_c J_c = \frac{a_c p}{\sqrt{2\pi mkT}} \tag{2.3-46}$$

where p is the partial pressure and m is the molecular mass. Using Eq. (2.3-46), and assuming $\alpha_c = 1$, the time that it takes to adsorb one monolayer can be estimated (refer to Table 2.3-6). The condensation coefficient depends on properties of the adsorbing molecule and of the surface. It also depends on the amount already adsorbed at the surface; it is often assumed that α_c is proportional to the surface fraction $(1-\theta)$ still available for adsorption: $\alpha_c = \alpha_0$ $(1-\theta)$. Also, chemical adsorption often is characterized by a potential barrier, which can be accounted for by assuming an Arrhenius behavior of α_c.

The rate of desorption, R_d, is proportional to the number of molecules adsorbed at the surface, N_c:

$$R_d = \frac{N_c}{\tau} = \frac{N_c}{\tau_0} e^{-Q/RT} \tag{2.3-47}$$

with R the universal gas constant and T the absolute temperature. The time constant τ describes the average residence time of the molecule at the surface, which depends on the period of vibration of the adsorbed molecule (typically $\tau_0 = 1 \times 10^{-13}$ s) and the bonding enthalpy Q. For physical adsorption $Q = 5-40$ kJ/mol while for chemical adsorption $Q = 40-800$ kJ/mol.

The net rate of change is given by Eqs. (2.3-46) and (2.3-47) as $R_c - R_d$. In equilibrium when $R_c - R_d = 0$, the concentration of adsorbed molecules can be calculated:

$$N_c = \frac{a_c p \tau_0}{\sqrt{2\pi mkT}} e^{-Q/RT} \tag{2.3-48}$$

2.4 SUMMARY

In this chapter a review of wafer contamination is presented. Different types of contaminants, as well as their sources and effects are discussed:

- Particle contamination is omnipresent, and indeed, is often generated by the manufacturing process itself. Its impact on the manufacturing process is mostly associated with the patterning process, and as such, it may be considered a "geometric," chemically inert defect.
- Metallic contamination may be present in different forms (e.g., molecular, particulate) and different specifications. Its impact usually is understood by its interaction (chemical or physical) with the device materials (e.g., silicon, dielectrics).
- Atmospheric molecular contamination, as the name implies, is airborne and has molecular dimensions. It covers a wide class of contaminants that can interact in different ways with the wafer surface.

Clearly this classification is a simplification and alternative interpretations are possible. The discussion in this chapter is aimed to be rather generic, in the sense that no explicit assumptions about processing conditions or device architecture are made. As device architectures develop, and as an increasing number of materials are being used ("More Moore" and "More than Moore"), the range of possible contaminants continues to increase. Understanding of the impact and behavior and contaminants therefore will remain a very relevant challenge for the years to come.

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